

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

## SZM DTC Troubleshooting Data

DOCUMENT VERSION: 1.4

RELEASE DATE: 03/05/24

| Author list |             |                |                       |                |                                             |
|-------------|-------------|----------------|-----------------------|----------------|---------------------------------------------|
| <i>Ver</i>  | <i>Date</i> | <i>Company</i> | <i>Author</i>         | <i>Contact</i> | <i>Telephone</i>                            |
| 1.4         | 03/05/24    | TML            | Body Electronics team | ---            | <a href="tel:020-66463576">020-66463576</a> |

|                                                                                  |                                                                                                        |                |
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|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

## Change History

| Ver. | Change details                                                                                    |
|------|---------------------------------------------------------------------------------------------------|
| 1.0  | <b>First Release</b> – DTC related information, Flowcharts for all features applicable added.     |
| 1.1  | <b>Second Release-</b> DTC U1000-00 and DTC U3000-49 Troubleshooting Information has been updated |
| 1.2  | <b>Third Release-</b> DTCs added as per review.                                                   |
| 1.3  | <b>Fourth Release-</b> DTCs B2233-01, B1F6B-11, B1F6A-71 updated as per review.                   |
| 1.4  | <b>Fifth Release-</b> DTCs B1F6A-13 updated as per review.                                        |

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|                                                                                  |                                                                                                        | Date: 03-05-24 |

## Abbreviation

| Name  | Abbreviation                             |
|-------|------------------------------------------|
| BCM   | Body control module                      |
| IPC   | Instrument Panel Cluster                 |
| HVAC  | Heating Ventilating and Air conditioning |
| SRS   | Safety restraint system                  |
| DTC   | Diagnostic trouble code                  |
| ABS   | Antilock braking system                  |
| VICM  | Vehicle immobilizer control module       |
| HU    | Head unit                                |
| CAN   | Controller area network                  |
| ECU   | Electronic control unit                  |
| N.A   | Not Applicable                           |
| OBD   | On Board diagnostic                      |
| CAN_H | CAN high                                 |
| CAN_L | CAN low                                  |
| ECM   | Engine control module                    |
| EPAS  | Electric power assistance system         |
| AFE   | Average fuel economy                     |
| IFE   | Instantaneous fuel economy               |
| MIL   | Malfunction indicator Lamp               |
| EBD   | Electronic brake force distribution      |
| SZM   | Seat Zone Module                         |
| CE    | Check engine                             |

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
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## Preliminary Check

1. Check resistance in Diagnostic CAN pin no. 6 and 14. It should be 120ohm.
2. Check for short conditions between CAN\_H-CAN\_L, CAN\_H-GND, CAN\_L-GND, CAN\_H-Vbatt. There should not be any short circuit conditions.
3. Check continuity between IPC CAN\_H (pin no. 15) vehicle side connector to OBD connector CAN\_H (pin no. 6) and CAN\_L (pin no. 14) vehicle side connector to ODB connector (pin no.14).
4. Turn ignition ON and Read IVN DTCs in IPC.

## Caution

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

## Notes

1. When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
2. Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
3. Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
4. If Diagnostic Trouble Code(s) are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.

DTC\_Code, Node 1

**B1087 – 88 :- LIN Bus A - bus off**

Text, Node 1

**Overview:**

Table, Node 1

|                            |                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------|
| ISO Code                   | B1087-88                                                                               |
| Customer Symptom           | Seat Movement Stops                                                                    |
| Fault Effects (On Vehicle) | Seat Movement Stops.<br>No communication between SZM and BCM                           |
| Lamp Status (If Any)       | N.A                                                                                    |
| Fault Detection Condition  | Seat Movement will Stop.                                                               |
| Normal Operating Condition | LIN A Communication works fine.                                                        |
| Probable Trouble Area      | 1. LIN transceiver fault.<br>2. BCM connector wired wrong.<br>3. Harness wire shorted. |
| Healing Condition          | 1. LIN communication established properly.<br>2. Clear DTC Command                     |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

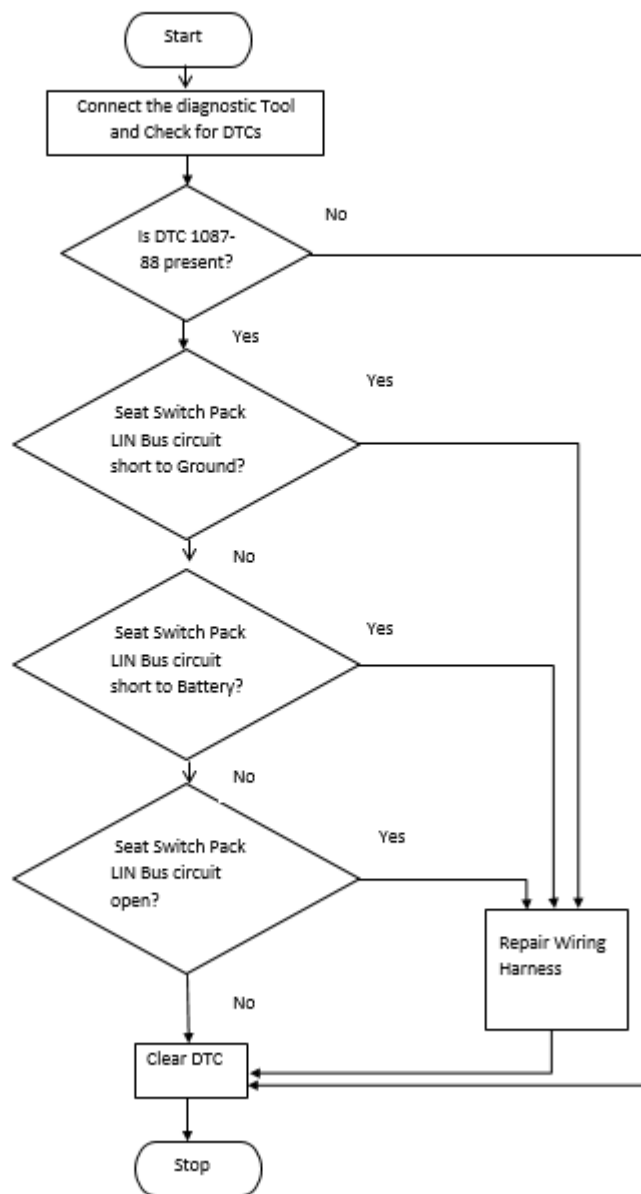
Date: 03-05-24

Text, Node 1

**Trouble Shooting: -**

Image, Node 1

B1087-88 LIN Bus A-Bus Off







**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 1

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 2

**B155A – 01:- General Electrical Failure of the motor or its Hall effect sensor failure or sensor's harness issues.(Slide Motor)**

Text, Node 2

**Overview:**

Table, Node 2

|                            |                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155A-01                                                                                                                                                                           |
| Customer Symptom           | Seat forward/reward –Slide movement stops                                                                                                                                          |
| Fault Effects (On Vehicle) | Seat Slide Movement Stops.                                                                                                                                                         |
| Lamp Status (If Any)       | N. A                                                                                                                                                                               |
| Fault Detection Condition  | Seat forward/reward –Slide movement stops                                                                                                                                          |
| Normal Operating Condition | Seat Slide movement works fine                                                                                                                                                     |
| Probable Trouble Area      | 1.Seat forward/reward motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2.Seat forward/reward motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Install new Slide motor<br>3. Clear DTC                                                                                                             |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

Version: 1.4

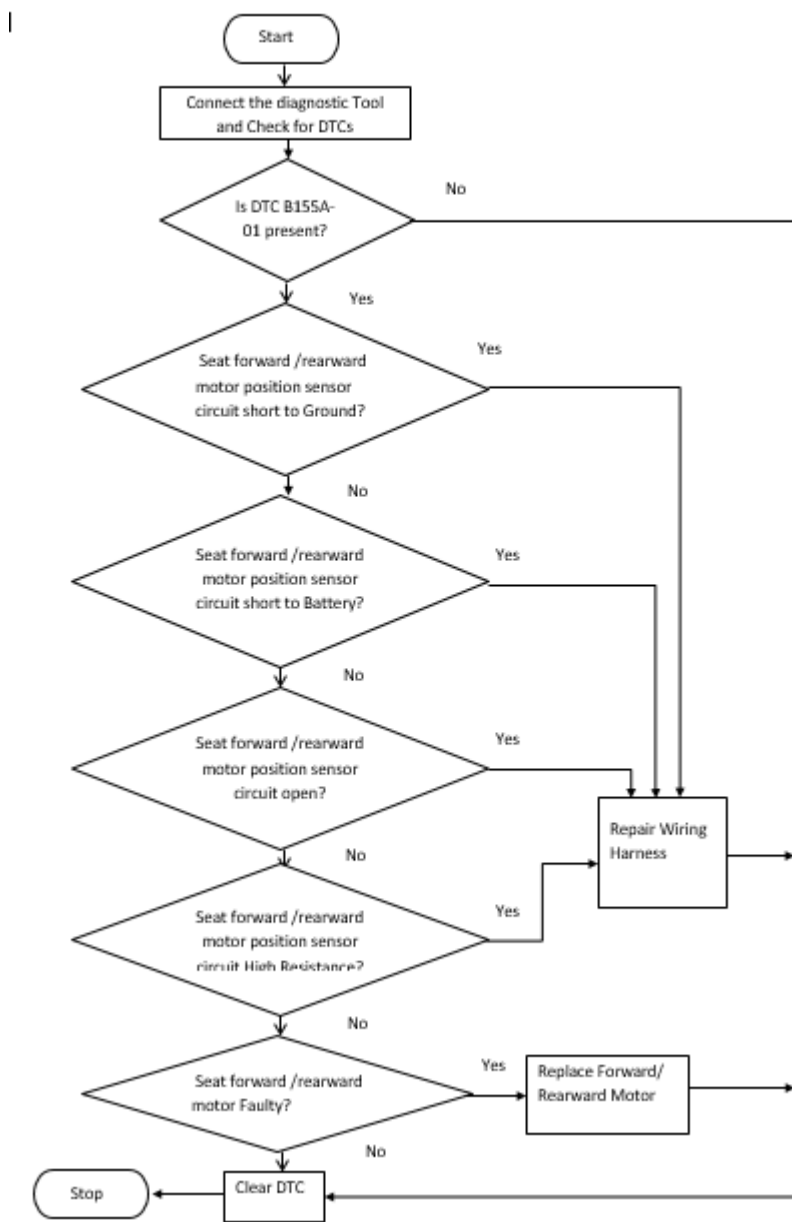
Date: 03-05-24

Text, Node 2

Trouble Shooting

Image, Node 2

## B155A-01 Seat Slide Motor Drive - General electrical failure



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 2

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- b. DTC should not be present in ECU memory.

**Circuit Schematic Diagram:-**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 3

**B155A-11:- Slide Motor Output Circuit Short To Ground**

Text, Node 3

**Overview:**

Table, Node 3

|                            |                                                                             |
|----------------------------|-----------------------------------------------------------------------------|
| ISO Code                   | B155A-11                                                                    |
| Customer Symptom           | Seat forward/reward –Slide movement stops                                   |
| Fault Effects (On Vehicle) | Seat Slide Movement Stops.                                                  |
| Lamp Status (If Any)       | N.A                                                                         |
| Fault Detection Condition  | Seat forward/reward –Slide movement stops                                   |
| Normal Operating Condition | Seat Slide movement works fine                                              |
| Probable Trouble Area      | 1.Seat forward/reward motor position sensor circuit short circuit to ground |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                    |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

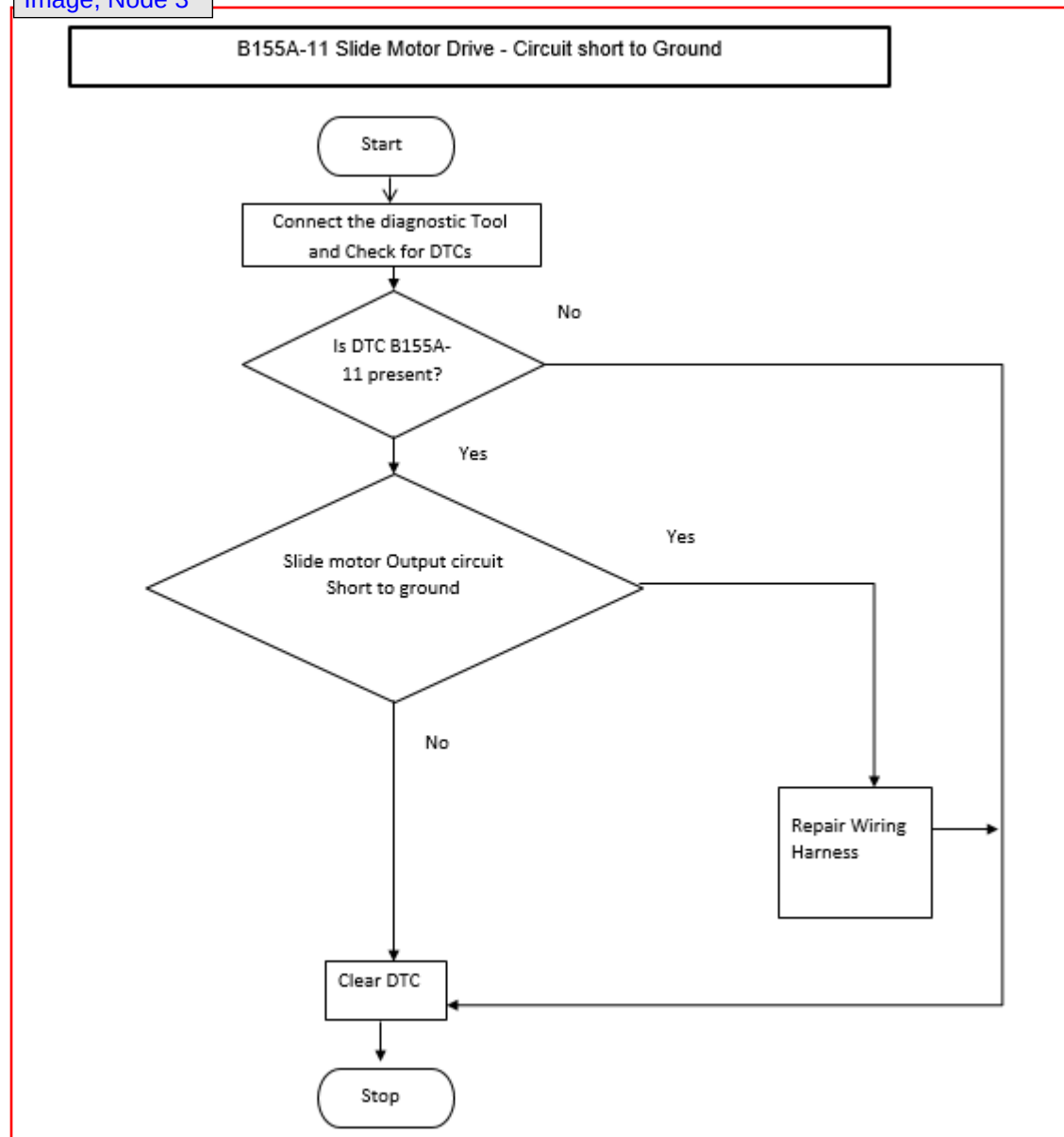
Version: 1.4

Date: 03-05-24

Text, Node 3

**Trouble Shooting: -**

Image, Node 3



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 3

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- c. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 4

**B155A-12:- Slide Motor Output Circuit Short To Battery**

Text, Node 4

**Overview:**

Table, Node 4

|                            |                                                                              |
|----------------------------|------------------------------------------------------------------------------|
| ISO Code                   | B155A-12                                                                     |
| Customer Symptom           | Seat forward/reward –Slide movement stops                                    |
| Fault Effects (On Vehicle) | Seat Slide Movement Stops.                                                   |
| Lamp Status (If Any)       | N.A                                                                          |
| Fault Detection Condition  | Seat forward/reward –Slide movement stops                                    |
| Normal Operating Condition | Seat Slide movement works fine                                               |
| Probable Trouble Area      | 1.Seat forward/reward motor position sensor circuit short circuit to Battery |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                     |





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

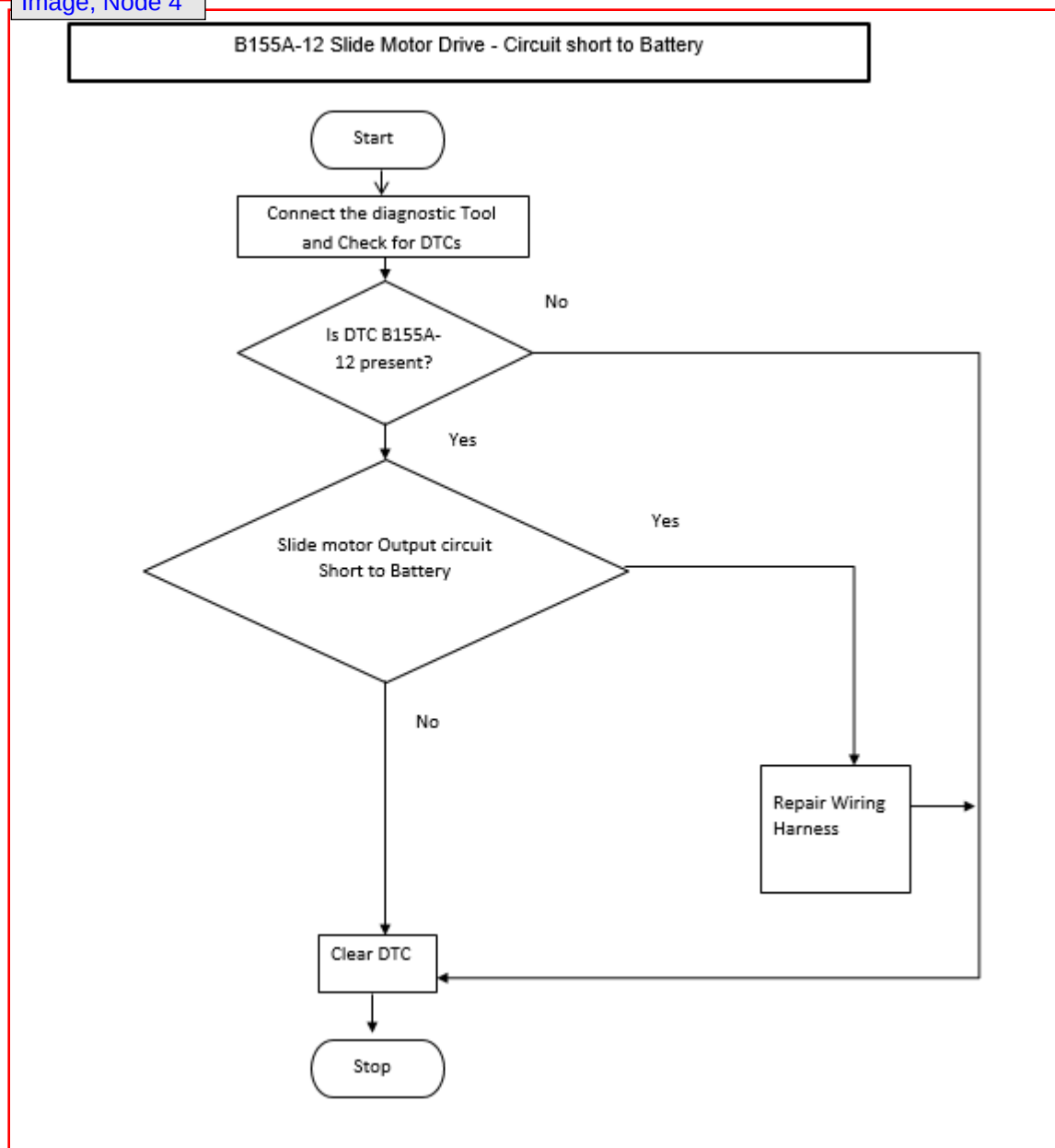
Version: 1.4

Date: 03-05-24

Text, Node 4

**Trouble Shooting:**

Image, Node 4



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 4

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- d. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

DTC\_Code, Node 5

**B155A – 18:- Slide Motor Output Open Circuit**

Text, Node 5

**Overview:**

Table, Node 5

|                            |                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155A-18                                                                                                                                                                           |
| Customer Symptom           | Seat forward/reward –Slide movement stops                                                                                                                                          |
| Fault Effects (On Vehicle) | Seat Slide Movement Stops.                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                |
| Fault Detection Condition  | Seat forward/reward –Slide movement stops                                                                                                                                          |
| Normal Operating Condition | Seat Slide movement works fine                                                                                                                                                     |
| Probable Trouble Area      | 1.Seat forward/reward motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2.Seat forward/reward motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Install new Slide motor<br>3. Clear DTC                                                                                                             |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

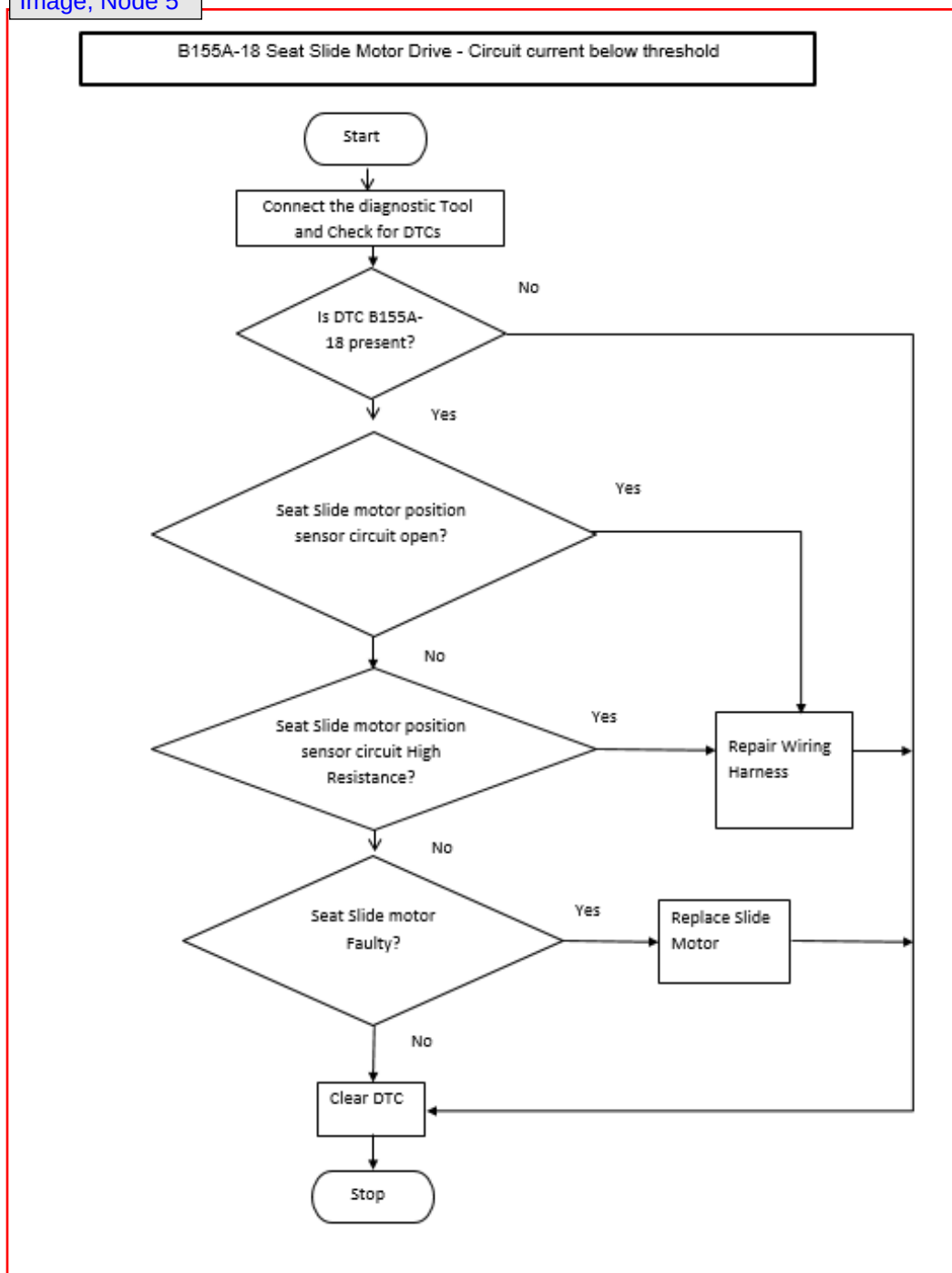
Version: 1.4

Date: 03-05-24

Text, Node 5

**Trouble Shooting: -**

Image, Node 5



Text, Node 5

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- e. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 6

### **B155A – 19:- Slide Motor Output Circuit Current Above Threshold**

Text, Node 6

#### **Overview:**

Table, Node 6

|                            |                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155A-19                                                                                                                                                                           |
| Customer Symptom           | Seat forward/reward –Slide movement stops                                                                                                                                          |
| Fault Effects (On Vehicle) | Seat Slide Movement Stops.                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                |
| Fault Detection Condition  | Seat forward/reward –Slide movement stops                                                                                                                                          |
| Normal Operating Condition | Seat Slide movement works fine                                                                                                                                                     |
| Probable Trouble Area      | 1.Seat forward/reward motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2.Seat forward/reward motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Install new Slide motor<br>3. Clear DTC                                                                                                             |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

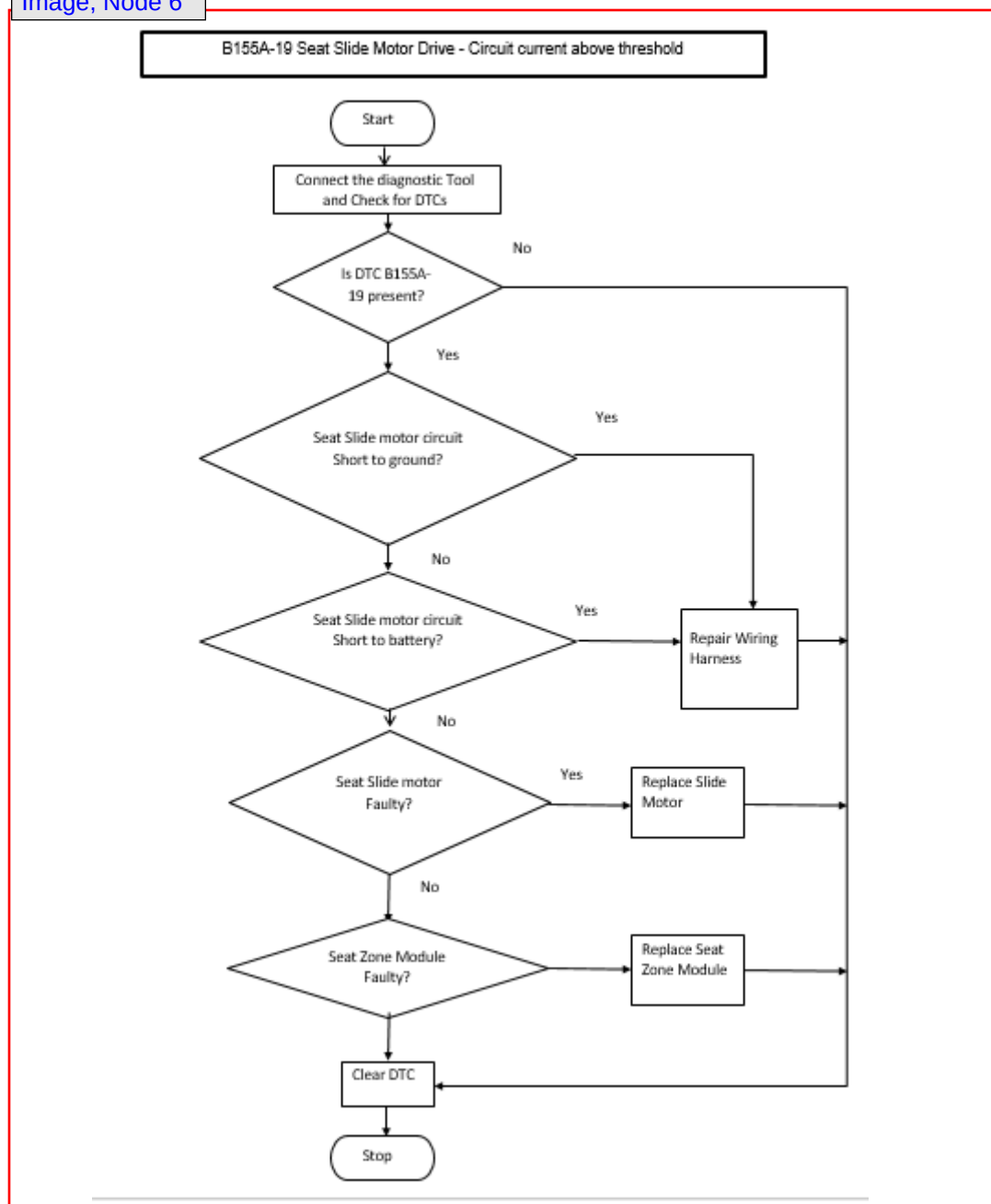
Version: 1.4

Date: 03-05-24

Text, Node 6

**Trouble Shooting: -**

Image, Node 6





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 6

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- f. DTC should not be present in ECU memory.

**Circuit Schematic Diagram:-**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



DTC\_Code, Node 7

**B155B – 01:- General Electrical Failure of the motor or its Hall effect sensor failure or sensor's harness issues.(Back rest Motor)**

Text, Node 7

**Overview:**

Table, Node 7

|                            |                                                                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155B-01                                                                                                                                                                                                                         |
| Customer Symptom           | Seat Squab Movement Stops                                                                                                                                                                                                        |
| Fault Effects (On Vehicle) | Seat Squab Movement Stops.                                                                                                                                                                                                       |
| Lamp Status (If Any)       | N. A                                                                                                                                                                                                                             |
| Fault Detection Condition  | Seat Squab Movement Stops.                                                                                                                                                                                                       |
| Normal Operating Condition | Seat Squab Movement works fine                                                                                                                                                                                                   |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Seat backrest motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>2. Seat backrest motor internal failure</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair wiring Harness</li> <li>2. Install new backrest motor</li> <li>3. Clear DTC</li> </ol>                                                                                          |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

Version: 1.4

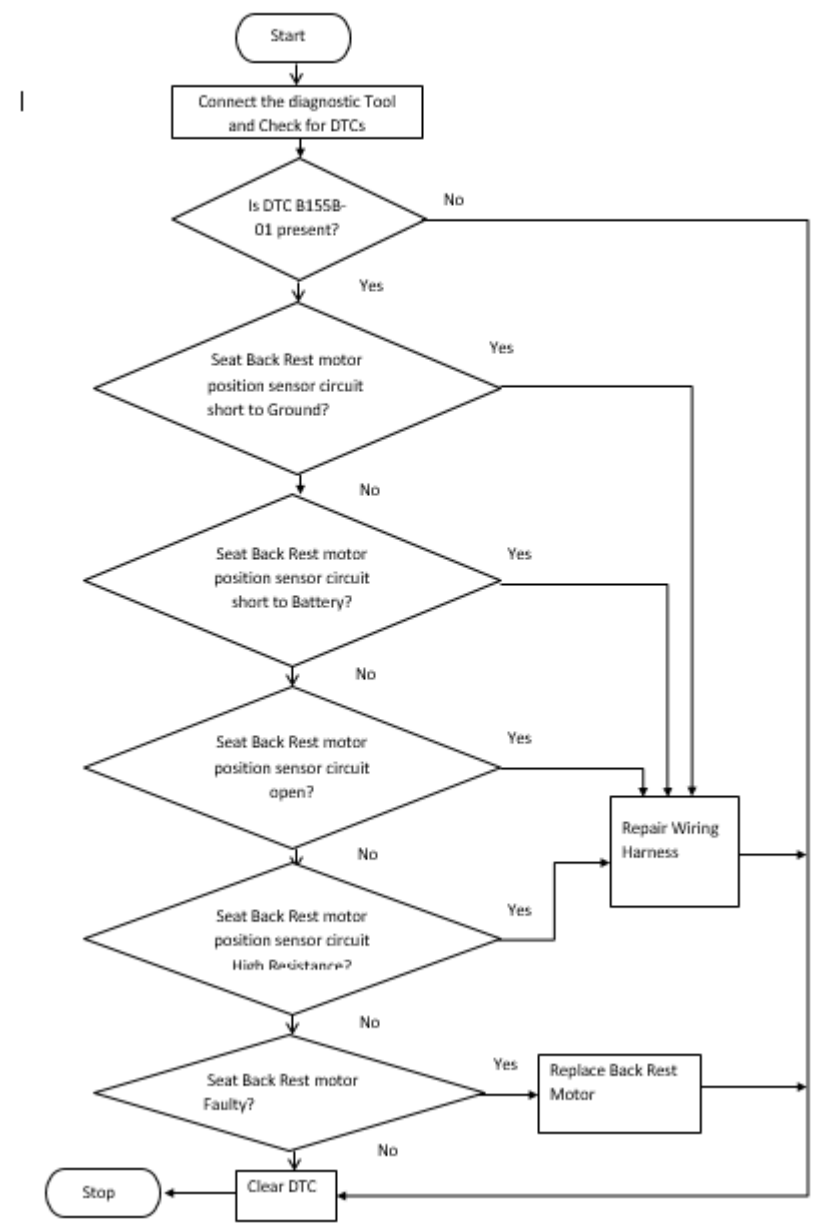
Date: 03-05-24

Text, Node 7

Trouble Shooting: -

Image, Node 7

## B155B-01 Seat Back Rest Motor Drive - General electrical failure



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 7

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- g. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

DTC\_Code, Node 8

**B155B-11 :- Back Rest Motor Output Circuit Short To Ground**

Text, Node 8

**Overview:**

Table, Node 8

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| ISO Code                   | B155B-11                                                                |
| Customer Symptom           | Seat Squab Movement Stops                                               |
| Fault Effects (On Vehicle) | Seat Squab Movement Stops.                                              |
| Lamp Status (If Any)       | N.A                                                                     |
| Fault Detection Condition  | Seat Squab Movement Stops.                                              |
| Normal Operating Condition | Seat Squab Movement works fine                                          |
| Probable Trouble Area      | 1. Seat backrest motor position sensor circuit short circuit to ground, |
| Healing Condition          | 1. Repair wiring Harness<br>2. Clear DTC                                |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

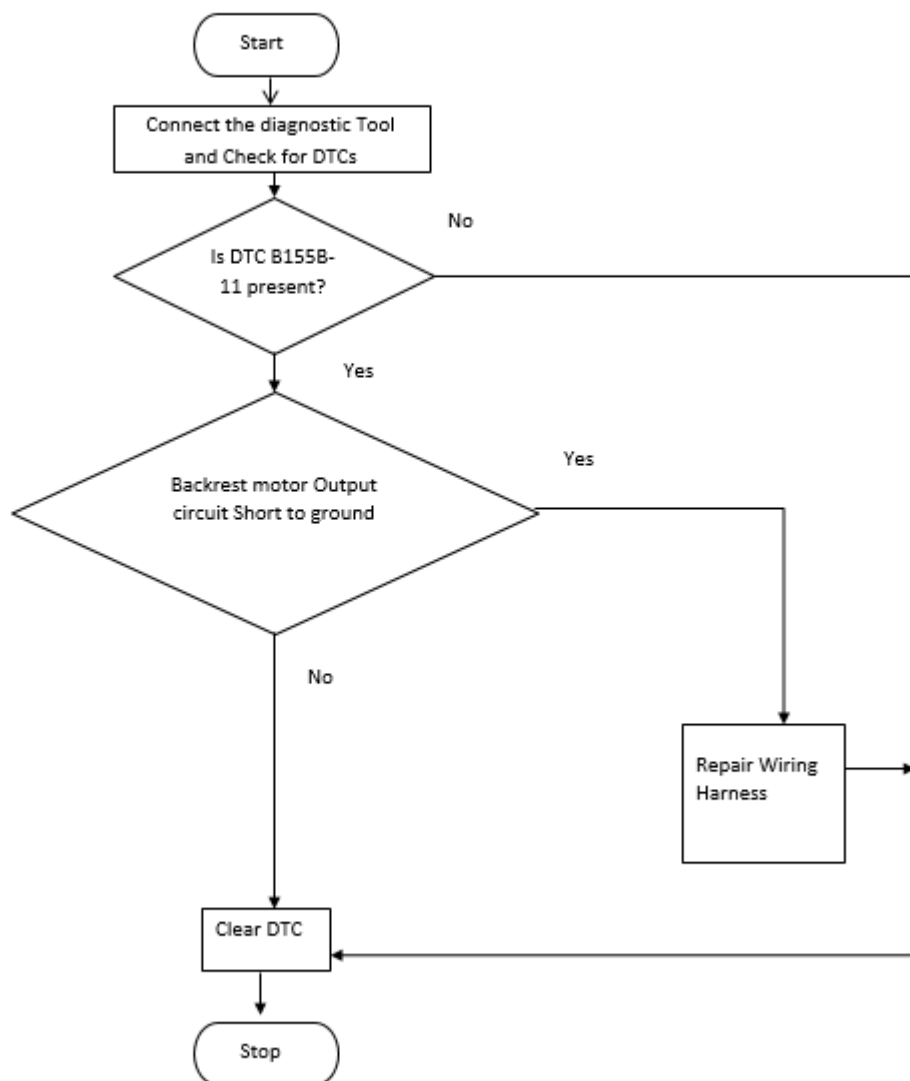
Date: 03-05-24

Text, Node 8

**Trouble Shooting: -**

Image, Node 8

**B155B-11 Backrest Slide Motor Drive - Circuit short to Ground**



Author: BCM COC

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|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 8

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- h. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 9

**B155B-12:- Back Rest Motor Output Circuit Short To Battery**

Text, Node 9

**Overview:**

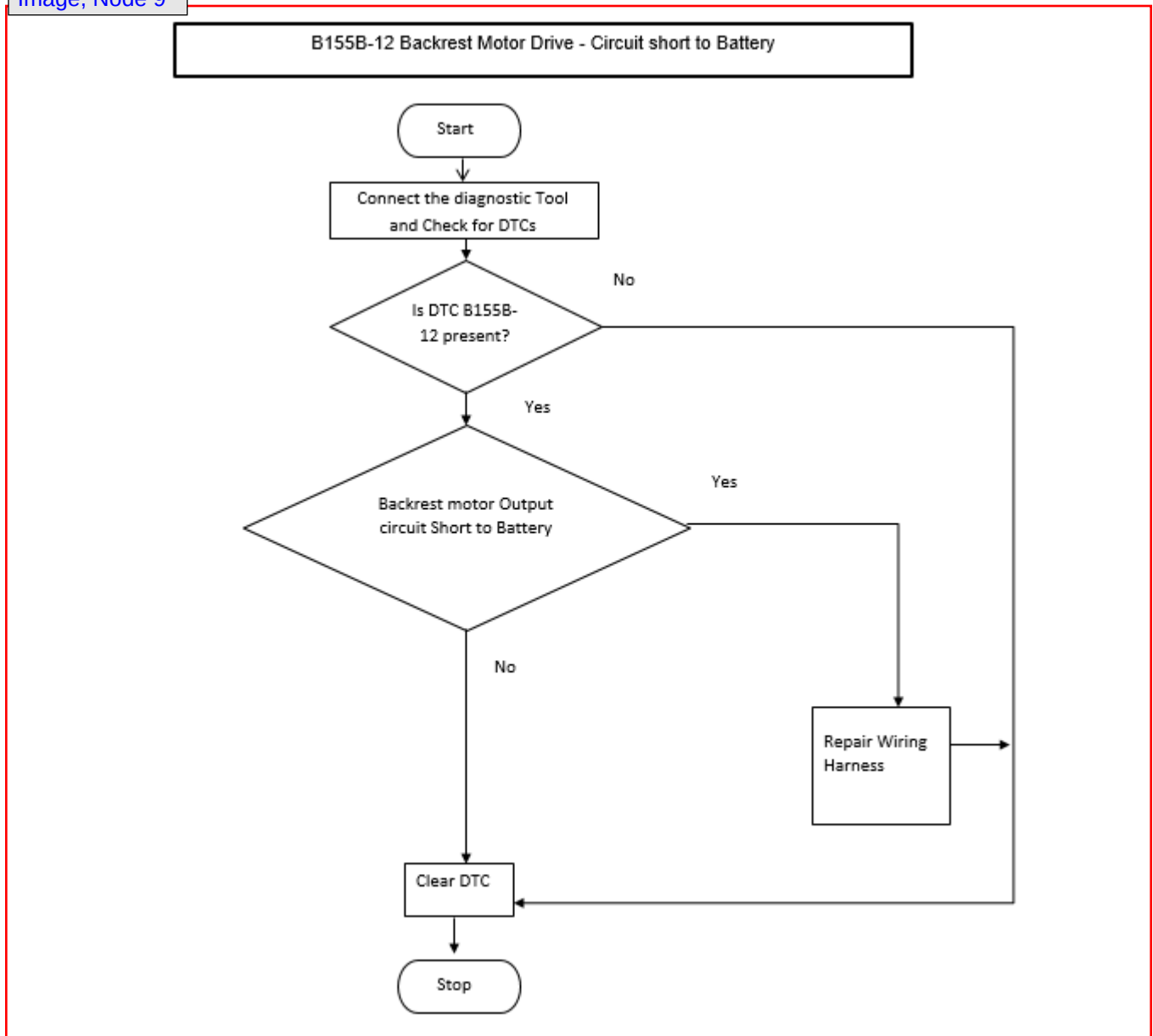
Table, Node 9

|                            |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| ISO Code                   | B155B-12                                                                |
| Customer Symptom           | Seat Squab Movement Stops                                               |
| Fault Effects (On Vehicle) | Seat Squab Movement Stops.                                              |
| Lamp Status (If Any)       | N. A                                                                    |
| Fault Detection Condition  | Seat Squab Movement Stops.                                              |
| Normal Operating Condition | Seat Squab Movement works fine                                          |
| Probable Trouble Area      | 1. Seat backrest motor position sensor circuit short circuit to battery |
| Healing Condition          | 1. Repair wiring Harness<br>2. Clear DTC                                |

Text, Node 9

**Trouble Shooting: -**

Image, Node 9





|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 9

#### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- i. DTC should not be present in ECU memory.

#### **Circuit Schematic Diagram: -**

N.A

#### **Component Location & Image: -**

N.A

#### **Connector Location: -**

N.A

#### **Connector View & Information: -**

N.A

#### **Inspection Method: -**

N.A



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 10

**B155B – 18:- Back Rest Motor Output Open Circuit**

Text, Node 10

**Overview:**

Table, Node 10

|                            |                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155B-18                                                                                                |
| Customer Symptom           | Seat Squab Movement Stops                                                                               |
| Fault Effects (On Vehicle) | Seat Squab Movement Stops.                                                                              |
| Lamp Status (If Any)       | N. A                                                                                                    |
| Fault Detection Condition  | Seat Squab Movement Stops.                                                                              |
| Normal Operating Condition | Seat Squab Movement works fine                                                                          |
| Probable Trouble Area      | 1. Seat backrest motor circuit open circuit, high resistance<br>2. Seat backrest motor internal failure |
| Healing Condition          | 1. Repair wiring Harness<br>2. Install new backrest motor<br>3. Clear DTC                               |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

Version: 1.4

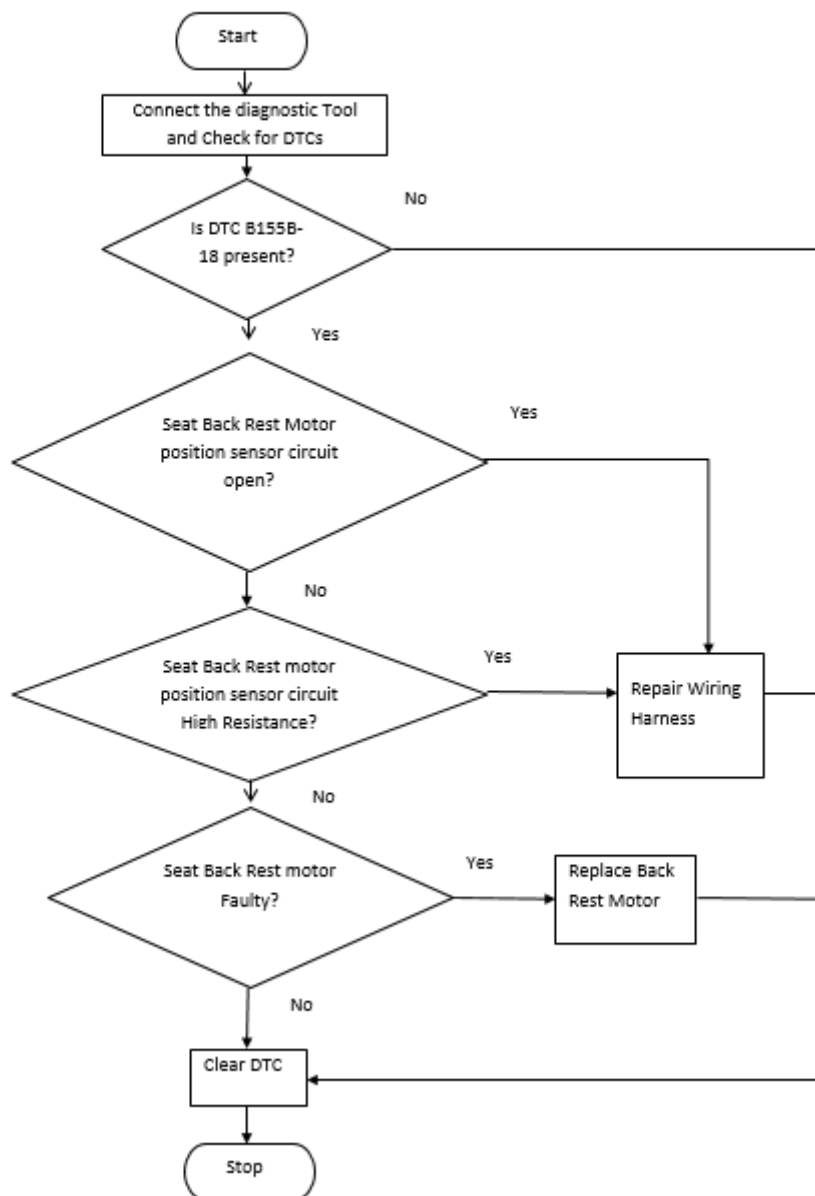
Date: 03-05-24

Text, Node 10

**Trouble Shooting: -**

Image, Node 10

B155B-18 Seat Back Rest Motor Drive - Circuit current below threshold



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 10

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- j. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 11

### **B155B – 19:- Back Rest Motor Output Circuit Current Above Threshold**

Text, Node 11

#### **Overview:**

Table, Node 11

|                            |                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155B-19                                                                                                                                                                          |
| Customer Symptom           | Seat Squab Movement Stops                                                                                                                                                         |
| Fault Effects (On Vehicle) | Seat Squab Movement Stops.                                                                                                                                                        |
| Lamp Status (If Any)       | N. A                                                                                                                                                                              |
| Fault Detection Condition  | Seat Squab Movement Stops.                                                                                                                                                        |
| Normal Operating Condition | Seat Squab Movement works fine                                                                                                                                                    |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Seat backrest motor circuit short circuit to ground, short circuit to power</li> <li>2. Seat backrest motor internal failure</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair wiring Harness</li> <li>2. Install new backrest motor</li> <li>3. Clear DTC</li> </ol>                                           |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

Version: 1.4

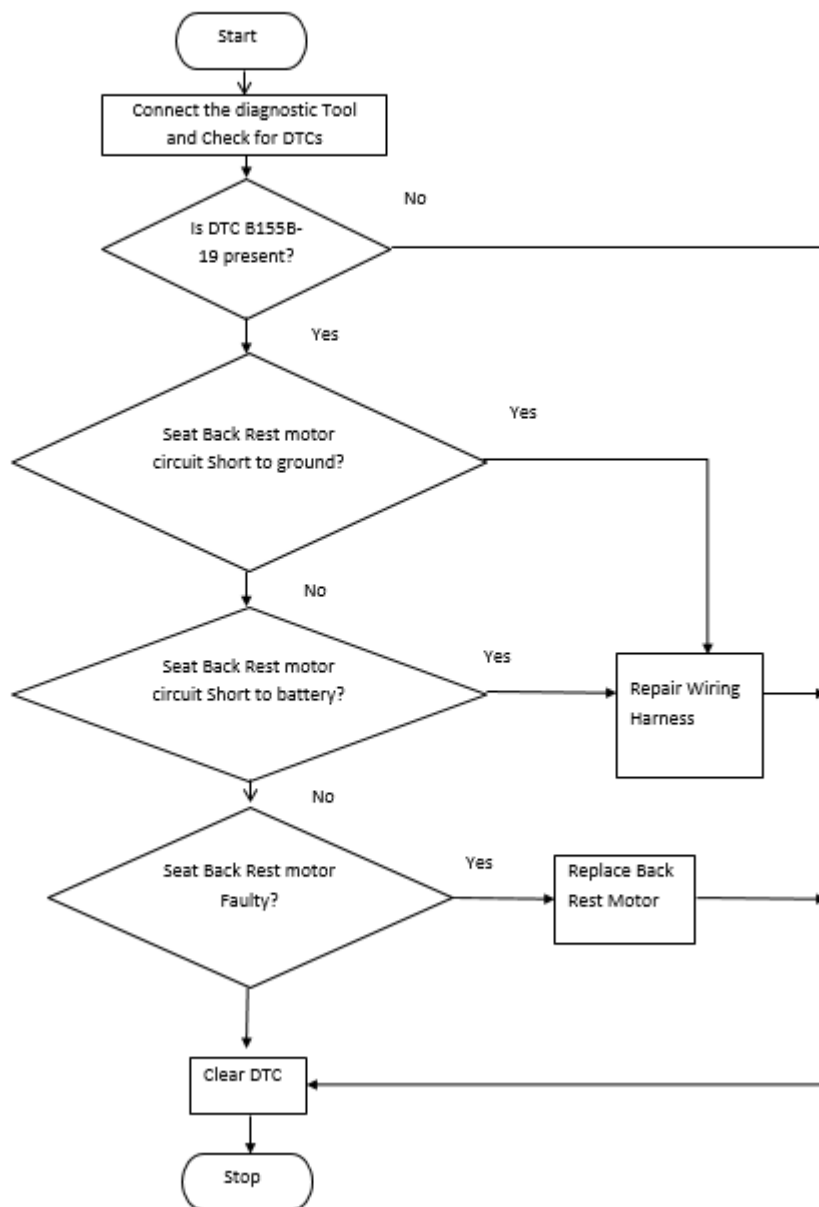
Date: 03-05-24

Text, Node 11

Trouble Shooting:

Image, Node 11

B155B-19 Seat Back Rest Motor Drive - Circuit current above threshold



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 11

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- k. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 12

**B1561 – 01:- General Electrical Failure of the motor or its Hall effect sensor failure or sensor's harness issues.(Height Motor)**

Text, Node 12

### Overview:

Table, Node 12

|                            |                                                                                                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1561-01                                                                                                                                                                                                                     |
| Customer Symptom           | Seat Height Movement Stops                                                                                                                                                                                                   |
| Fault Effects (On Vehicle) | Seat Height Movement Stops                                                                                                                                                                                                   |
| Lamp Status (If Any)       | N. A                                                                                                                                                                                                                         |
| Fault Detection Condition  | Seat Height Movement Stops                                                                                                                                                                                                   |
| Normal Operating Condition | Seat Height Movement Works Fine                                                                                                                                                                                              |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Seat height motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>2. Seat height motor internal failure</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair Wiring Harness</li> <li>2. Install Height Motor</li> <li>3. Clear DTC</li> </ol>                                                                                            |





# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

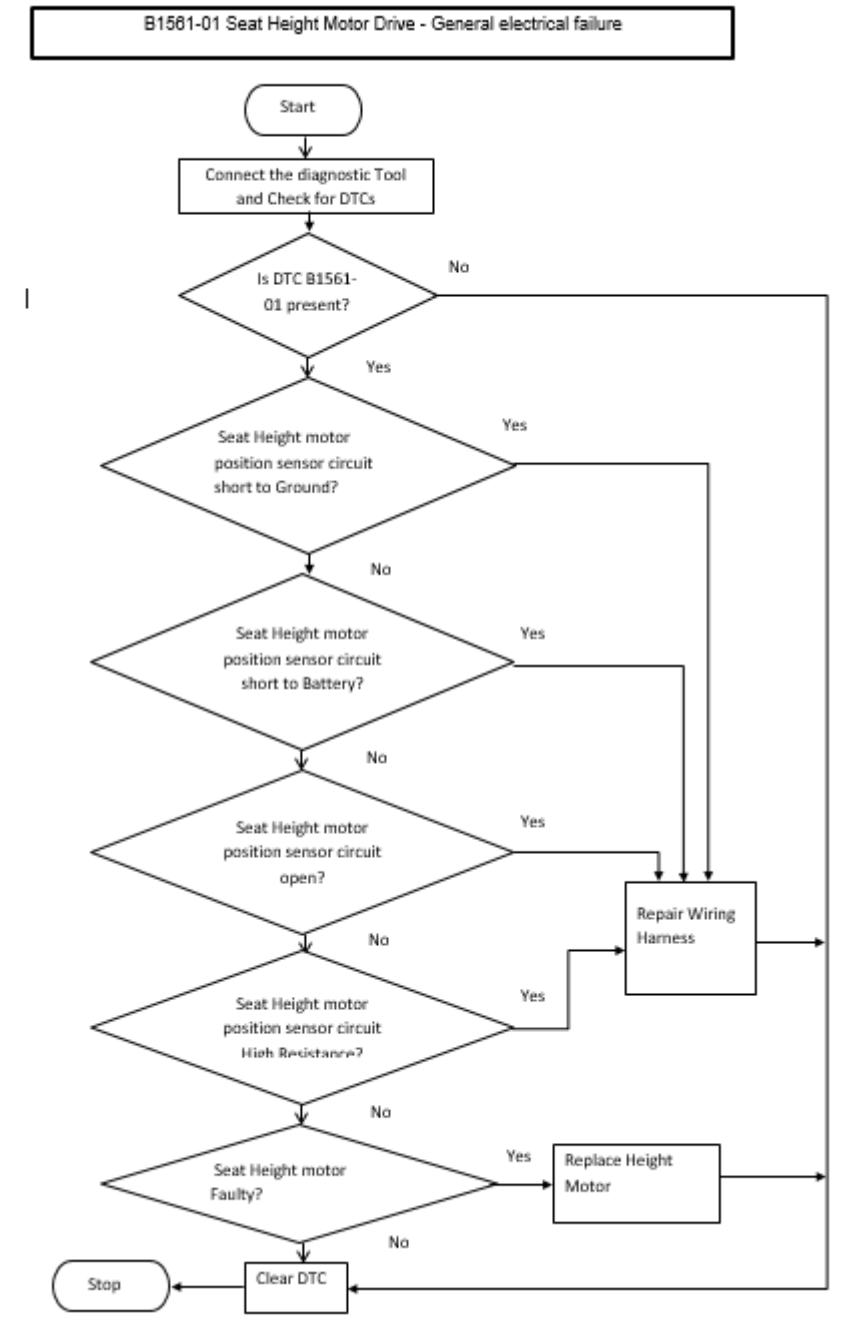
Version: 1.4

Date: 03-05-24

Text, Node 12

**Trouble Shooting: -**

Image, Node 12



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 12

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- I. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A

DTC\_Code, Node 13

**B1561-11 :- Height Motor Output Circuit Short To Ground**

Text, Node 13

**Overview:**

Table, Node 13

|                            |                                                                      |
|----------------------------|----------------------------------------------------------------------|
| ISO Code                   | B1561-11                                                             |
| Customer Symptom           | Seat Height Movement Stops                                           |
| Fault Effects (On Vehicle) | Seat Height Movement Stops                                           |
| Lamp Status (If Any)       | N. A                                                                 |
| Fault Detection Condition  | Seat Height Movement Stops                                           |
| Normal Operating Condition | Seat Height Movement Works Fine                                      |
| Probable Trouble Area      | 1. Seat height motor position sensor circuit short circuit to ground |
| Healing Condition          | 1. Repair Wiring Harness<br>2. Clear DTC                             |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

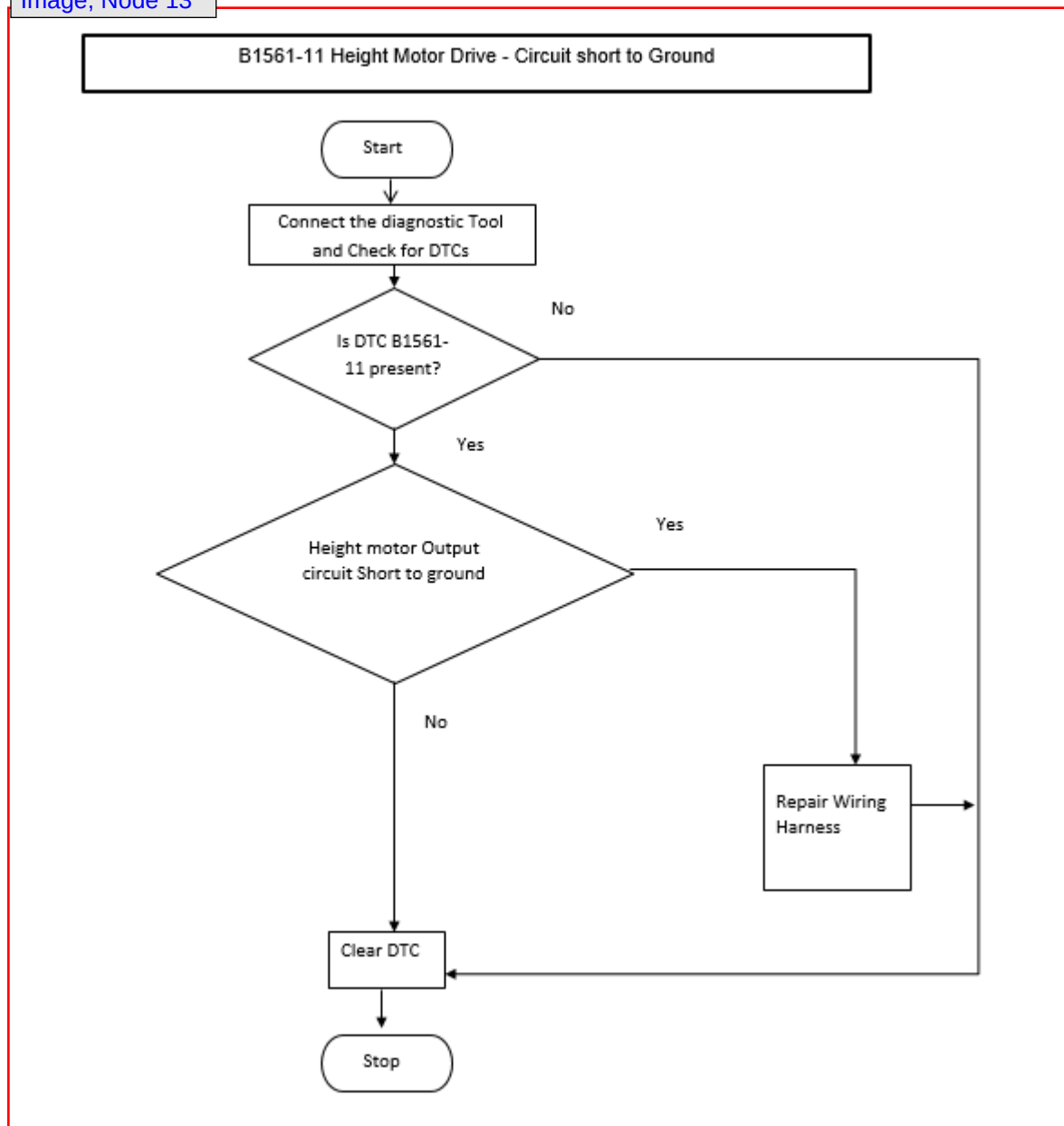
Version: 1.4

Date: 03-05-24

Text, Node 13

**Trouble Shooting: -**

Image, Node 13



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 13

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- m. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

DTC\_Code, Node 14

**B1561-12:- Height Motor Output Circuit Short To Battery**

Text, Node 14

**Overview:**

Table, Node 14

|                            |                                                                       |
|----------------------------|-----------------------------------------------------------------------|
| ISO Code                   | B1561-12                                                              |
| Customer Symptom           | Seat Height Movement Stops                                            |
| Fault Effects (On Vehicle) | Seat Height Movement Stops                                            |
| Lamp Status (If Any)       | N. A                                                                  |
| Fault Detection Condition  | Seat Height Movement Stops                                            |
| Normal Operating Condition | Seat Height Movement Works Fine                                       |
| Probable Trouble Area      | 1. Seat height motor position sensor circuit short circuit to battery |
| Healing Condition          | 1. Repair Wiring Harness<br>2. Clear DTC                              |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

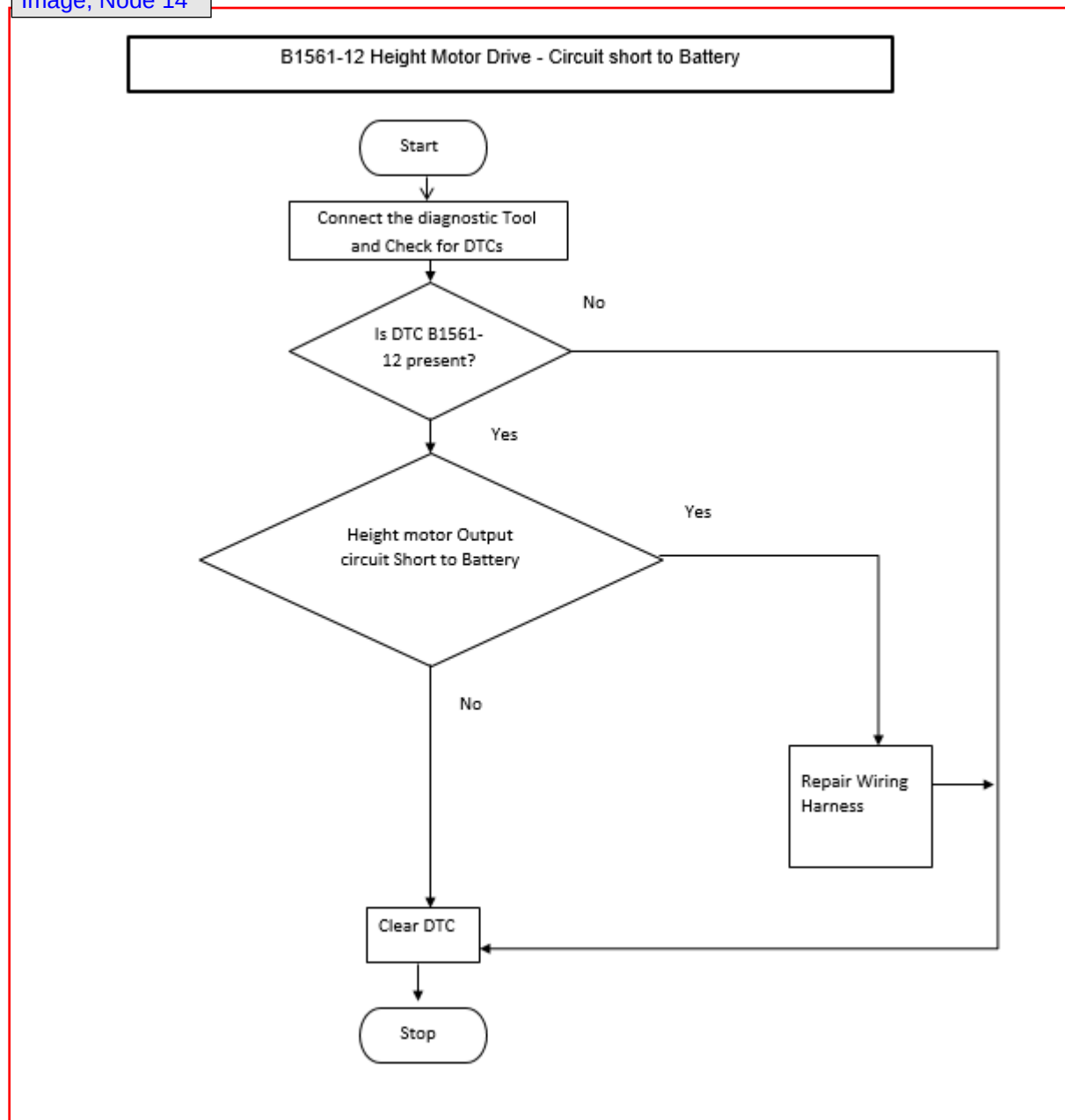
Version: 1.4

Date: 03-05-24

Text, Node 14

**Trouble Shooting: -**

Image, Node 14



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 14

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- n. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



DTC\_Code, Node 15

**B1561 – 18:- Height Motor Output Open Circuit**

Text, Node 15

**Overview:**

Table, Node 15

|                            |                                                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1561-18                                                                                                                                                    |
| Customer Symptom           | Seat Height Movement Stops                                                                                                                                  |
| Fault Effects (On Vehicle) | Seat Height Movement Stops                                                                                                                                  |
| Lamp Status (If Any)       | N.A                                                                                                                                                         |
| Fault Detection Condition  | Seat Height Movement Stops                                                                                                                                  |
| Normal Operating Condition | Seat Height Movement Works Fine                                                                                                                             |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Seat height motor circuit open circuit, high resistance</li> <li>2. Seat height motor internal failure</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair Wiring Harness</li> <li>2. Install Height Motor</li> <li>3. Clear DTC</li> </ol>                           |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

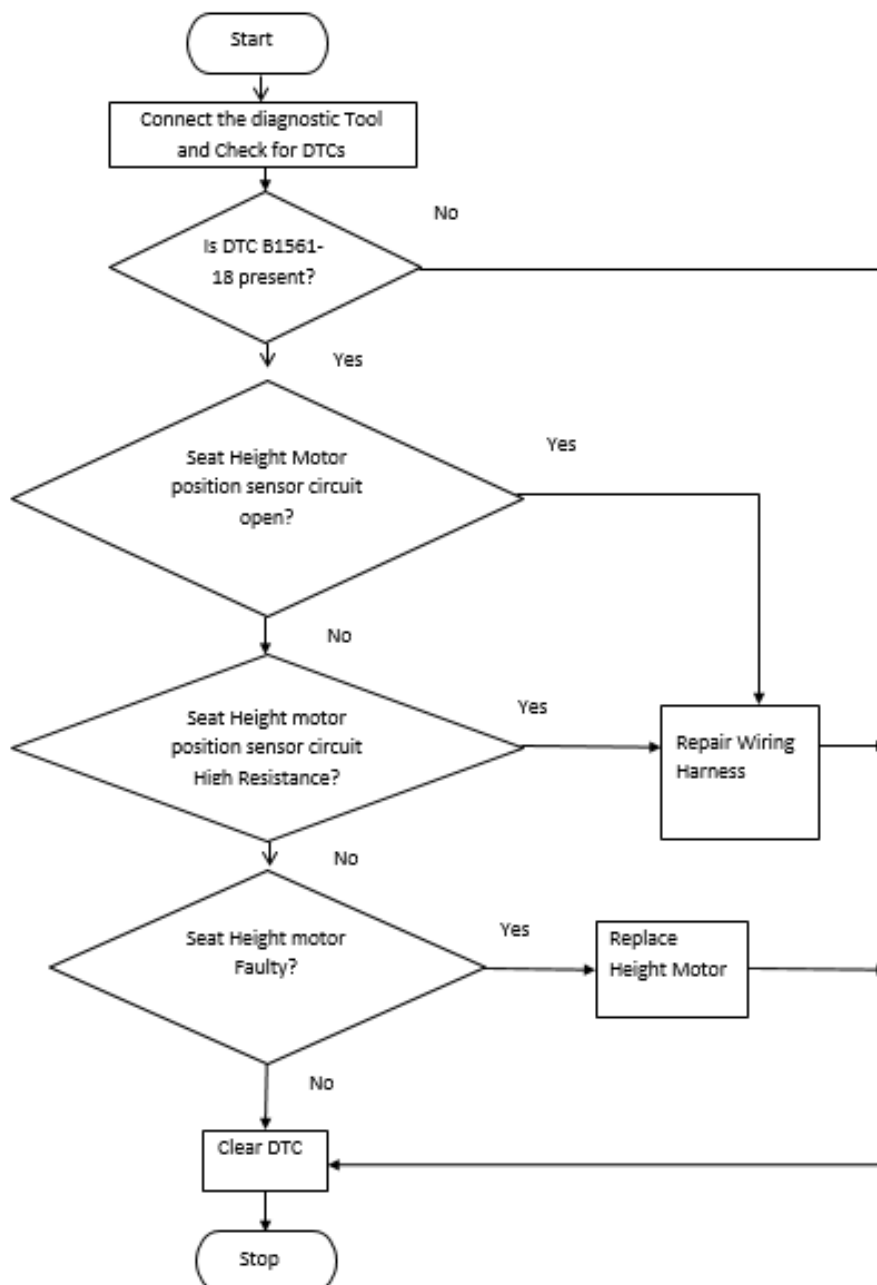
Date: 03-05-24

Text, Node 15

**Trouble Shooting:-**

Image, Node 15

B1561-18 Seat Height Motor Drive - Circuit current below threshold



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 15

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- o. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

DTC\_Code, Node 16

**B1561 – 19:- Height Motor Output Circuit Current Above Threshold**

Text, Node 16

**Overview:**

Table, Node 16

|                            |                                                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1561-19                                                                                                                                                                      |
| Customer Symptom           | Seat Height Movement Stops                                                                                                                                                    |
| Fault Effects (On Vehicle) | Seat Height Movement Stops                                                                                                                                                    |
| Lamp Status (If Any)       | N. A                                                                                                                                                                          |
| Fault Detection Condition  | Seat Height Movement Stops                                                                                                                                                    |
| Normal Operating Condition | Seat Height Movement Works Fine                                                                                                                                               |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Seat height motor circuit short circuit to ground, short circuit to power</li> <li>2. Seat height motor internal failure</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair Wiring Harness</li> <li>2. Install Height Motor</li> <li>3. Clear DTC</li> </ol>                                             |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

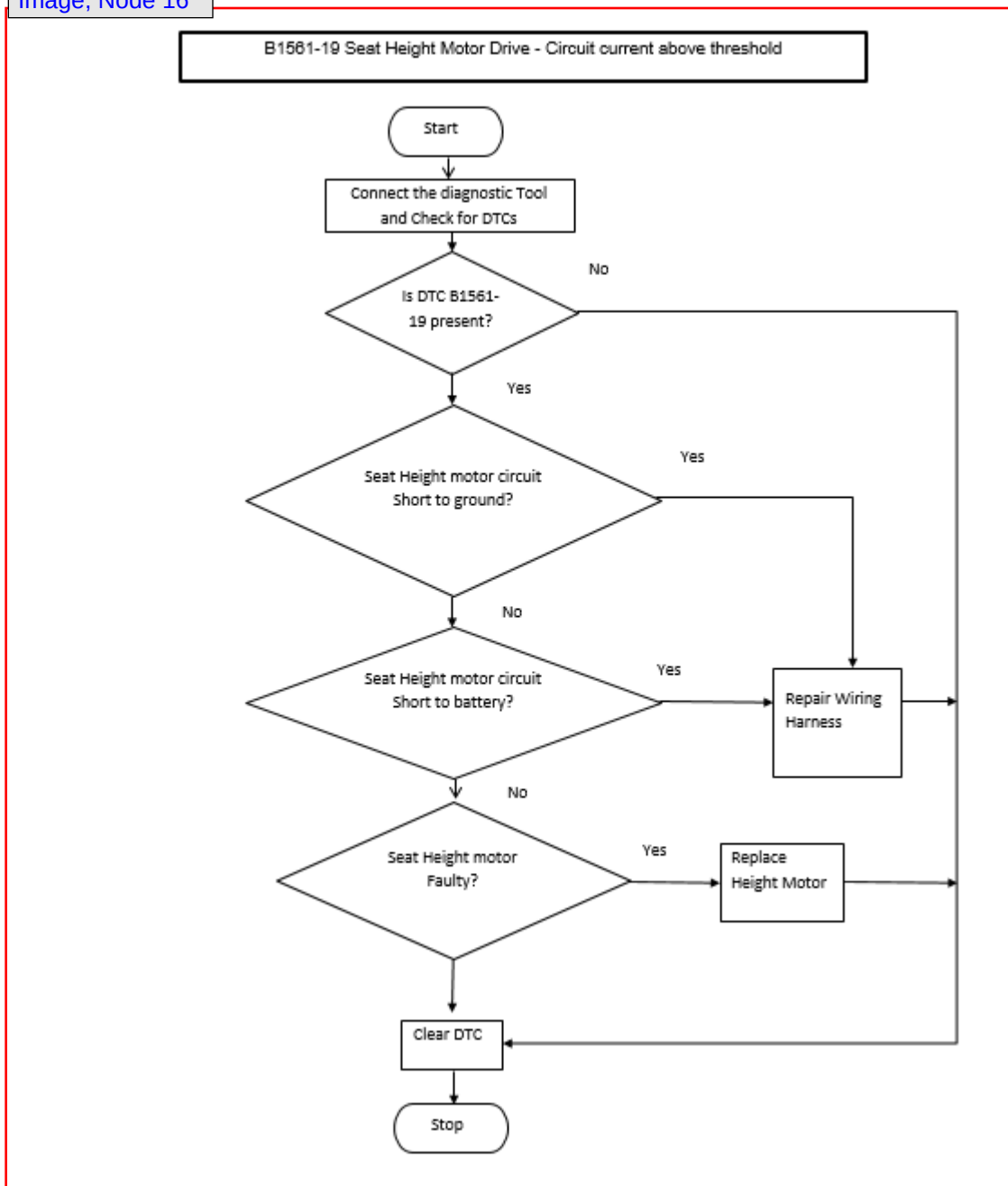
Version: 1.4

Date: 03-05-24

Text, Node 16

**Trouble Shooting: -**

Image, Node 16



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 16

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- p. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 17

### B1D1E – 24:- Switch Pack Switch stuck high

Text, Node 17

#### **Overview:**

Table, Node 17

|                            |                                                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1D1E-24                                                                                                                                                             |
| Customer Symptom           | Seat Switch Not Pressed/Switch Stuck                                                                                                                                 |
| Fault Effects (On Vehicle) | Seat Switch Not Pressed/Switch Stuck                                                                                                                                 |
| Lamp Status (If Any)       | N. A                                                                                                                                                                 |
| Fault Detection Condition  | Seat Switch Not Pressed/Switch Stuck                                                                                                                                 |
| Normal Operating Condition | Seat Switch getting Press/No stuck Condition                                                                                                                         |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Seat switch pack stuck active</li> <li>2. Seat switch pack internal failure</li> </ol>                                     |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Free Stuck Switches</li> <li>2. Install a new front seat switch pack as necessary</li> <li>3. Clear DTC Command</li> </ol> |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

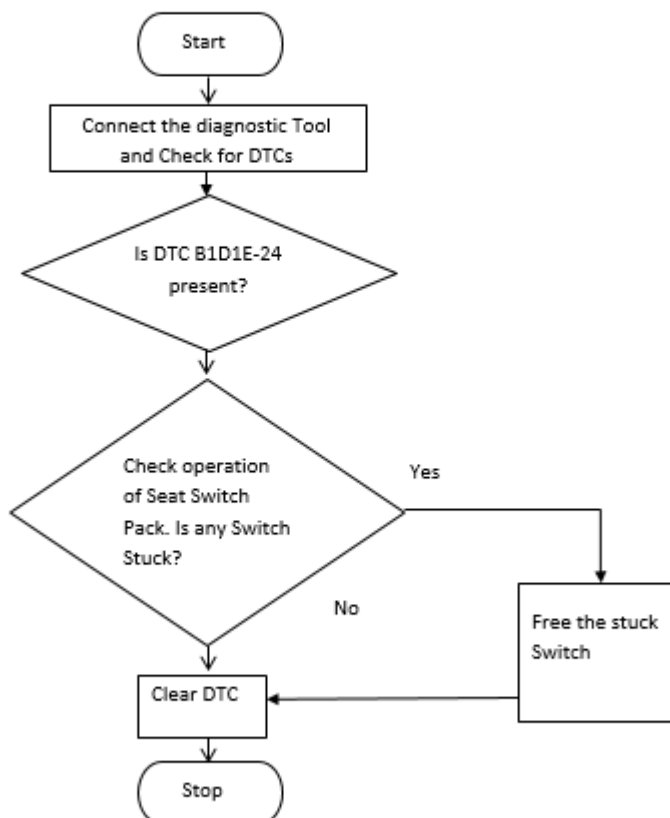
Date: 03-05-24

Text, Node 17

**Trouble Shooting: -**

Image, Node 17

**B1D1E-24 Seat Switch Pack - Signal stuck high**





|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 17

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- q. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A

DTC\_Code, Node 18

**B1D1E – 87:- Seat Switch Pack Missing Message**

Text, Node 18

**Overview:**

Table, Node 18

|                            |                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1D1E-87                                                                                                                                                                                                                                                                                         |
| Customer Symptom           | No seat Movement after pressing button from Switch Pack                                                                                                                                                                                                                                          |
| Fault Effects (On Vehicle) | <ol style="list-style-type: none"> <li>1. No seat Movement after pressing button from Switch Pack</li> <li>2. No communication between SZM and BCM</li> </ol>                                                                                                                                    |
| Lamp Status (If Any)       | N. A                                                                                                                                                                                                                                                                                             |
| Fault Detection Condition  | No seat Movement after pressing button from Switch Pack                                                                                                                                                                                                                                          |
| Normal Operating Condition | <ol style="list-style-type: none"> <li>1. After Pressing Switches/Buttons Seat movements works fine</li> <li>2. LIN A Communication works fine.</li> </ol>                                                                                                                                       |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. Front seat switch pack power or ground circuit open circuit, high resistance</li> <li>2. LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>3. Front seat switch pack internal failure</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair Wiring Harness</li> <li>2. Install new Seat Switch Pack</li> <li>3. Clear DTC</li> </ol>                                                                                                                                                        |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

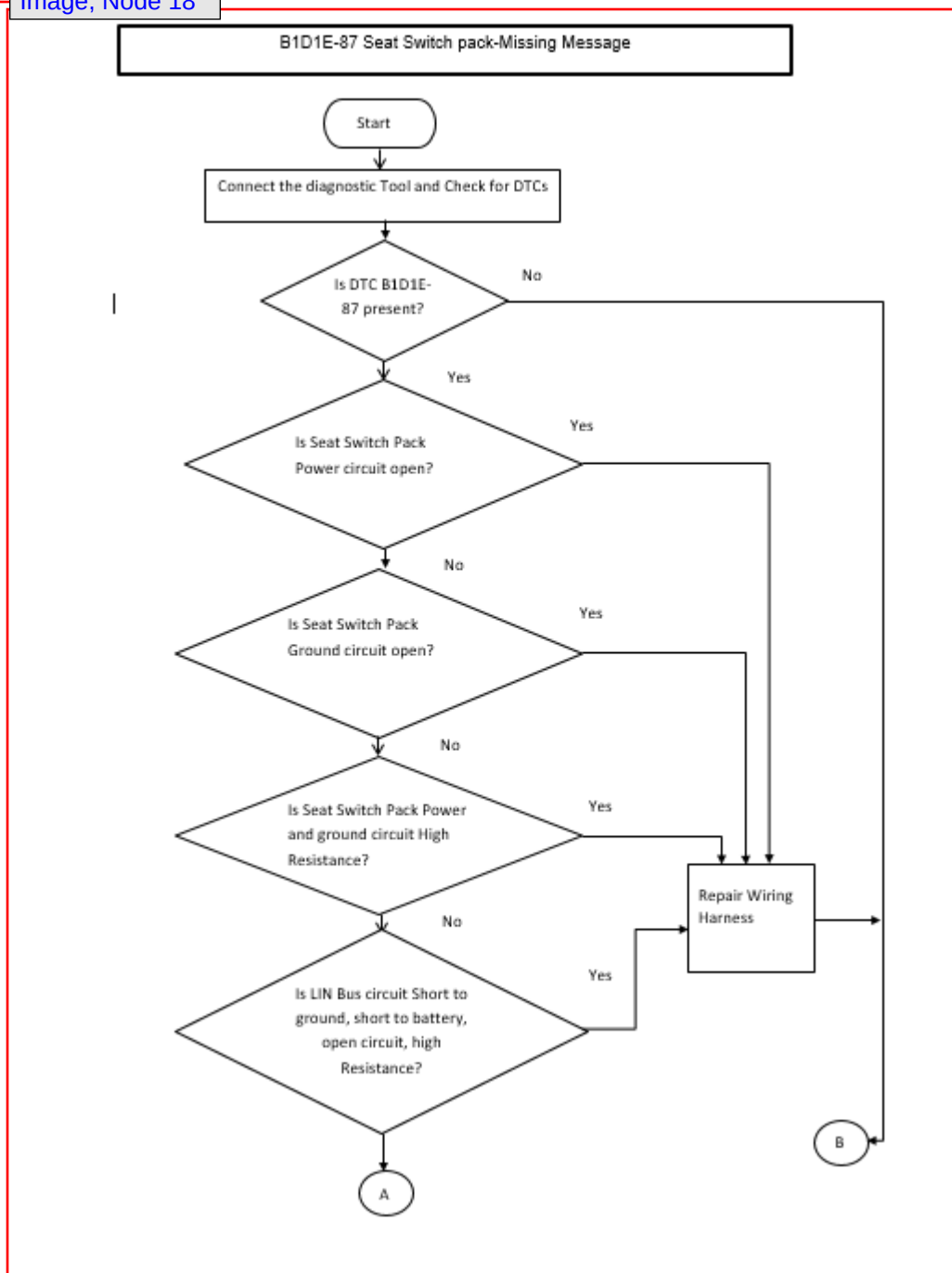
Version: 1.4

Date: 03-05-24

Text, Node 18

Trouble Shooting

Image, Node 18



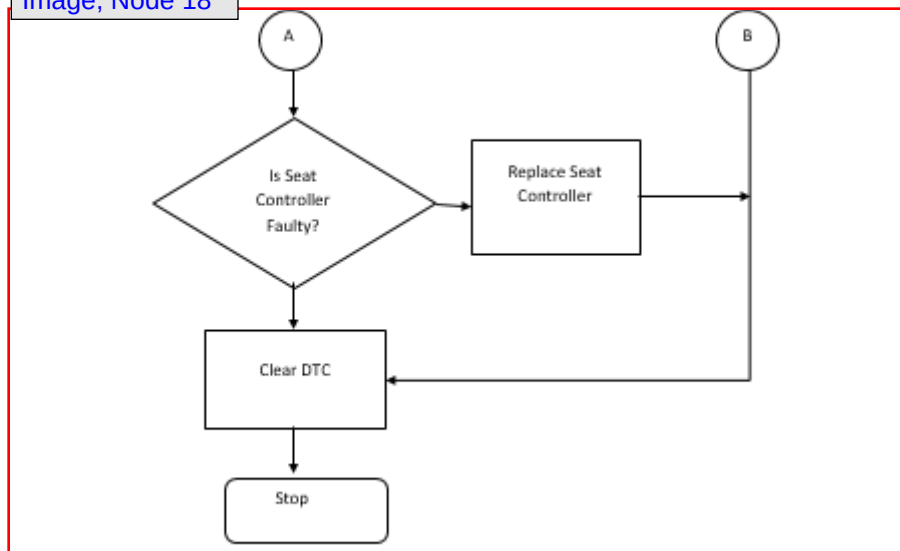


**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Image, Node 18



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 18

### DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- r. DTC should not be present in ECU memory.

### Circuit Schematic Diagram: -

N.A

### Component Location & Image: -

N.A

### Connector Location: -

N.A

### Connector View & Information: -

N.A

### Inspection Method: -

N.A

DTC Code, Node 19

**U1000-00: Solid State Driver Protection Active - Driver Disabled (Output driver is disabled temporarily)**

Text, Node 19

**Overview:**

Table, Node 19

|                            |                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | U1000-00                                                                                                                                   |
| Customer Symptom           | Seat Movement Stops                                                                                                                        |
| Fault Effects (On Vehicle) | Seat Movement Stops.                                                                                                                       |
| Lamp Status (If Any)       | N.A                                                                                                                                        |
| Fault Detection Condition  | Seat Movement Stops.                                                                                                                       |
| Normal Operating Condition | Seat Movement works fine.                                                                                                                  |
| Probable Trouble Area      | 1. Control module internal output driver is disabled due to a fault condition detected on an output circuit to protect the control module. |
| Healing Condition          | 1. Check the seat module for output circuit<br>2. Clear DTC Command                                                                        |

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 19

### **Trouble Shooting: -**

Text, Node 19

#### **NOTE:**

1. One of the output driver is disabled because it has experienced too many short to GND (Short To BATT in case of low side drives) and it has reached 1/3rd or 2/3rd of it life time. Another DTC will be logged to show which output has the fault.
2. The fault must be rectified on the output circuit and then only run the routine 040E as per routine sheet to clear the pin inhibit and then only clear the DTC.
3. Repeated settings of this diagnostic trouble code will result in unrecoverable control module shut down to prevent module damage
4. Check the seat module for output circuit Diagnostic Trouble Code(s) and perform the relevant corrective action

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 19

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- s. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 20

**U3000 – 93:-Control Module-SZM control module's Application software and Calibration are not matching.**

Text, Node 20

**Overview:**

Table, Node 20

|                            |                                                                 |
|----------------------------|-----------------------------------------------------------------|
| ISO Code                   | U3000-93                                                        |
| Customer Symptom           | No Seat Movement                                                |
| Fault Effects (On Vehicle) | No Seat Movement<br>No communication between SZM and BCM        |
| Lamp Status (If Any)       | N.A                                                             |
| Fault Detection Condition  | No Seat Movement                                                |
| Normal Operating Condition | Seat Movement works fine                                        |
| Probable Trouble Area      | Application and calibration mismatch with vehicle specification |
| Healing Condition          | Clear DTC Command                                               |

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 20

### **Trouble Shooting: -**

- Using the diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
- If the fault persists, contact TML technical assistance

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- t. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4<br><br>Date: 03-05-24 |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------|

DTC\_Code, Node 21

**U3000 – 49:- Control Module-Output driver is disabled permanently**

Text, Node 21

**Overview:**

Table, Node 21

|                            |                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | U3000-49                                                                                                                                        |
| Customer Symptom           | No Seat Movement                                                                                                                                |
| Fault Effects (On Vehicle) | No Seat Movement                                                                                                                                |
| Lamp Status (If Any)       | N.A                                                                                                                                             |
| Fault Detection Condition  | No Seat Movement                                                                                                                                |
| Normal Operating Condition | LIN A Communication works fine.                                                                                                                 |
| Probable Trouble Area      | 1. Seat module internal failure due to an output circuit short circuit to ground, short circuit to power<br><br>2. Seat module internal failure |
| Healing Condition          | 1. Repair Wiring Harness<br>2. Install New Seat Module<br>3. Clear DTC Command                                                                  |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 21

**Trouble Shooting: -**

**NOTE:**

1. This DTC is set when one or more short circuits have been detected. The relevant output circuits have been disabled permanently
2. One of the output driver has experienced too many short to GND (Short To BATT in case of low side drives) and it has reached its life time and has been permanently disabled to protect the ECU. Another DTC will be logged to show which output has the fault. Its a permanent fault and the ECU has to be replaced.
3. Using the Jaguar Land Rover approved diagnostic equipment, check for other short circuit related Diagnostic Trouble Code(s) and perform the relevant corrective actions. If the fault persists, install a new seat module

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- u. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 22

**B160F- 19:- LIN Power Feed-Circuit Current Above Threshold**

Text, Node 22

**Overview:**

Table, Node 22

|                            |                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| ISO Code                   | B160F19                                                                                        |
| Customer Symptom           | No Seat Movement                                                                               |
| Fault Effects (On Vehicle) | No Seat Movement<br>No communication between SZM and BCM                                       |
| Lamp Status (If Any)       | N.A                                                                                            |
| Fault Detection Condition  | No Seat Movement                                                                               |
| Normal Operating Condition | LIN A Communication works fine.                                                                |
| Probable Trouble Area      | 1. Lin power feed circuit short circuit to ground                                              |
| Healing Condition          | 2. LIN communication established properly.<br>3. Repair Wiring Harness<br>4. Clear DTC Command |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

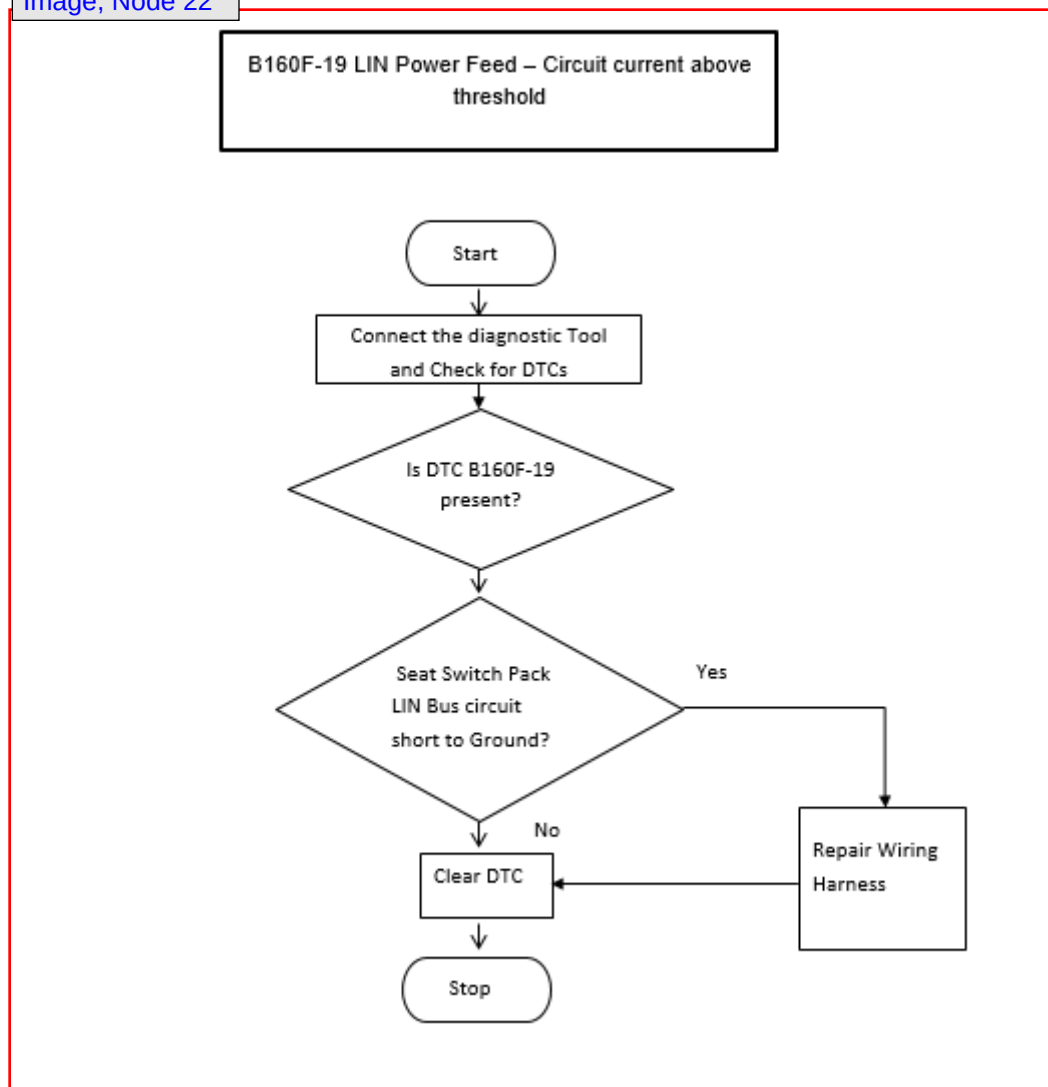
Version: 1.4

Date: 03-05-24

Text, Node 22

**Trouble Shooting: -**

Image, Node 22





# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

Version: 1.4

Date: 03-05-24

Text, Node 22

## DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- v. DTC should not be present in ECU memory.

## Circuit Schematic Diagram: -

N.A

## Component Location & Image: -

N.A

## Connector Location: -

N.A

## Connector View & Information: -

N.A

## Inspection Method: -

N.A



DTC\_Code, Node 23

**B2233 - 13:- ORVM Co-Driver horizontal axis motor Open Circuit**

Text, Node 23

**Overview:**

Table, Node 23

|                            |                                                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2233-13                                                                                                                                                                                      |
| Customer Symptom           | ORVM Co-Driver horizontal axis motor stops                                                                                                                                                    |
| Fault Effects (On Vehicle) | ORVM Co-Driver mirror stops                                                                                                                                                                   |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                           |
| Fault Detection Condition  | ORVM Co-Driver mirror stops                                                                                                                                                                   |
| Normal Operating Condition | ORVM Co-Driver mirror stops works                                                                                                                                                             |
| Probable Trouble Area      | 1.ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Co-Driver horizontal axis motor</u> internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Install new motor<br>3. Clear DTC                                                                                                                              |





# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

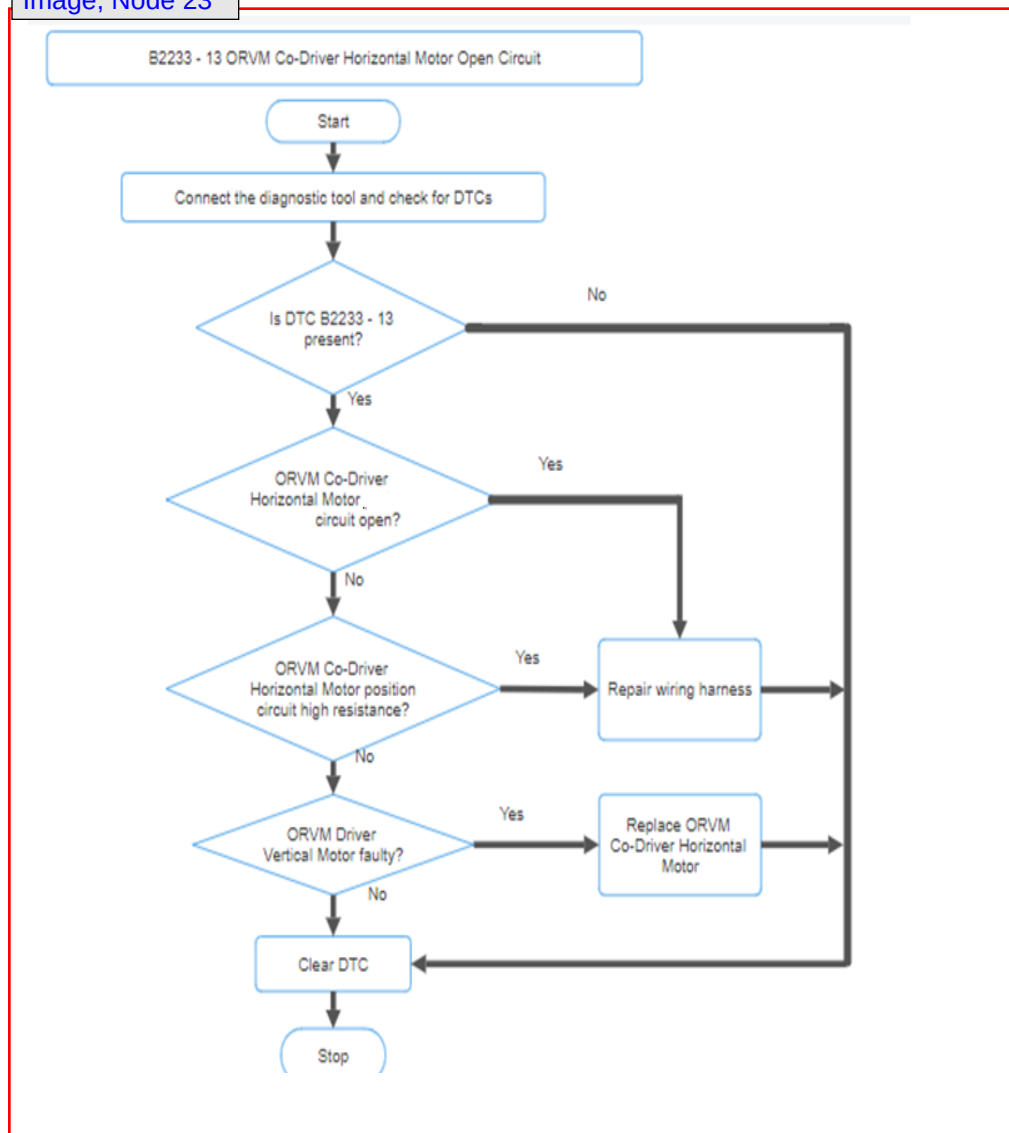
Version: 1.4

Date: 03-05-24

Text, Node 23

**Trouble Shooting: -**

Image, Node 23



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 23

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A



DTC\_Code, Node 24

**B2233-11:- ORVM Co-Driver horizontal axis motor STG**

Text, Node 24

**Overview:**

Table, Node 24

|                            |                                                                      |
|----------------------------|----------------------------------------------------------------------|
| ISO Code                   | B2233-11                                                             |
| Customer Symptom           | ORVM Co-driver horizontal axis motor movement stops                  |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops.                                |
| Lamp Status (If Any)       | N.A                                                                  |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                 |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                            |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                             |





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

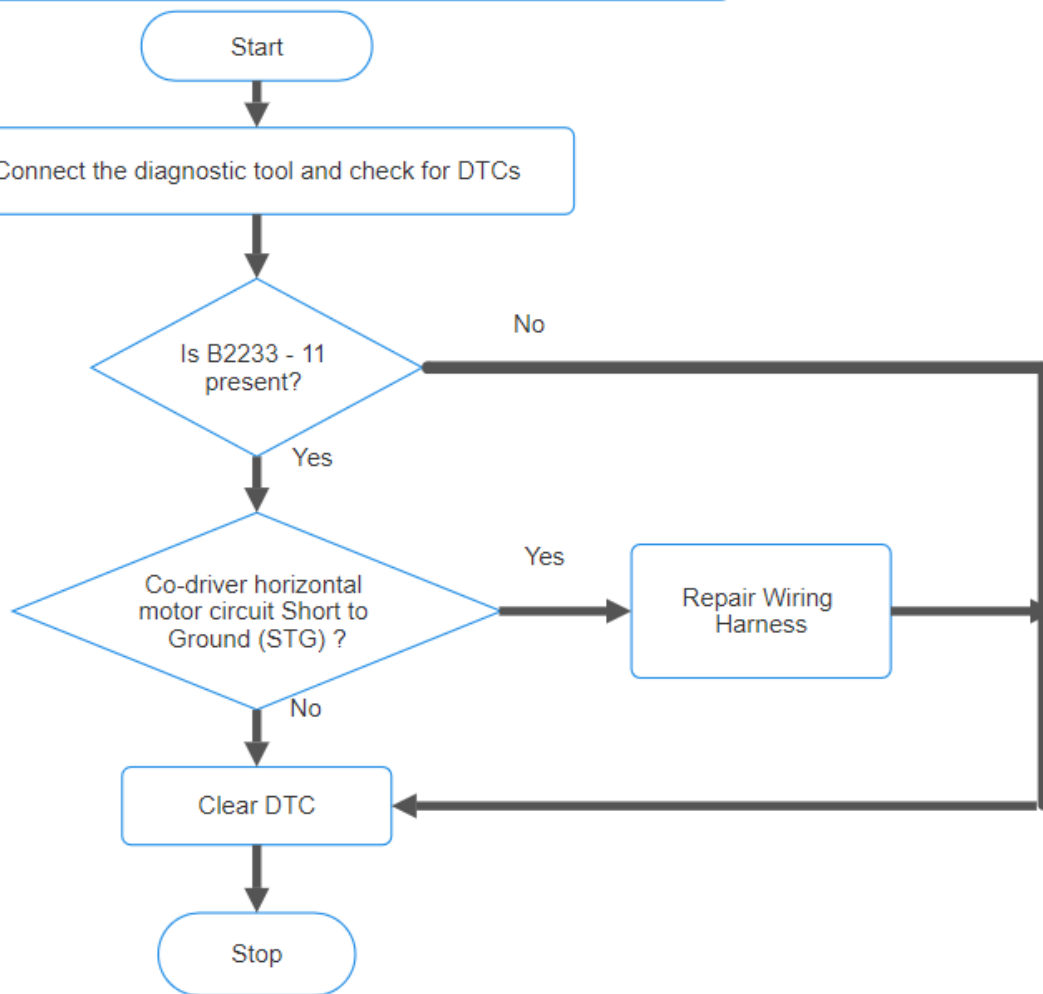
Date: 03-05-24

Text, Node 24

**Trouble Shooting: -**

Image, Node 24

B2233 - 11 ORVM Co-Driver Horizontal Motor Circuit Short to Ground (STG)



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 24

#### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

#### **Circuit Schematic Diagram: -**

N.A

#### **Component Location & Image: -**

N.A

#### **Connector Location: -**

N.A

#### **Connector View & Information: -**

N.A

#### **Inspection Method: -**

N.A





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 25

**B2233-12:- ORVM Co-Driver horizontal axis motor STB**

Text, Node 25

**Overview:**

Table, Node 25

|                            |                                                                       |
|----------------------------|-----------------------------------------------------------------------|
| ISO Code                   | B2233-12                                                              |
| Customer Symptom           | S ORVM Co-driver horizontal axis motor movement stops                 |
| Fault Effects (On Vehicle) | ORVM Co-driver horizontal axis motor Stops.                           |
| Lamp Status (If Any)       | N.A                                                                   |
| Fault Detection Condition  | ORVM Co-driver horizontal axis motor movement stops                   |
| Normal Operating Condition | ORVM Co-driver horizontal axis motor works fine                       |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to Battery |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                              |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

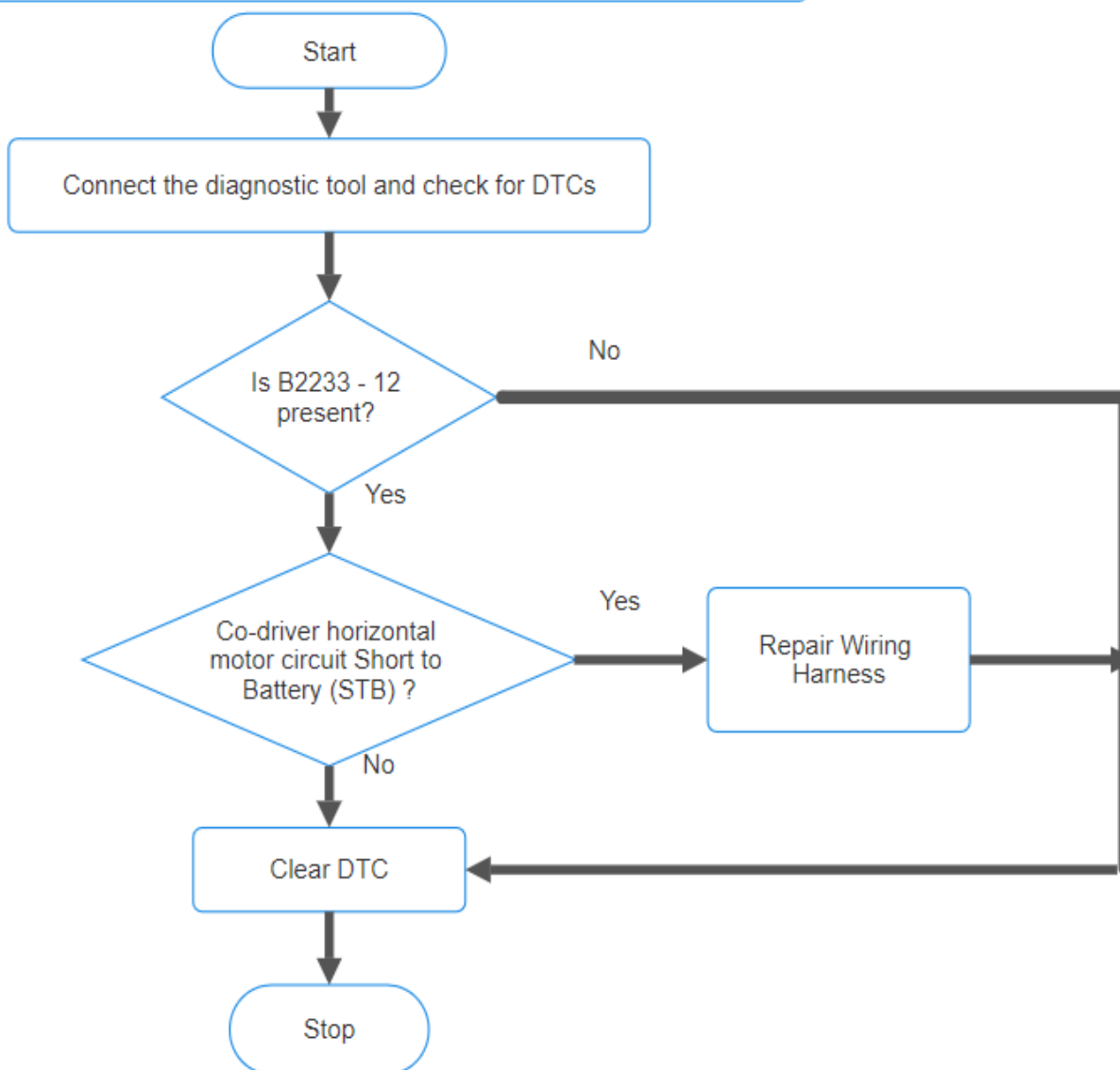
Date: 03-05-24

Text, Node 25

**Trouble Shooting: -**

Image, Node 25

B2233 - 12 ORVM Co-Driver Horizontal Motor Circuit Short to Battery (STB)



Author: BCM COC

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|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 25

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A





DTC\_Code, Node 26

**B2233-19:- ORVM Co-Driver horizontal axis motor Over current**

Text, Node 26

**Overview:**

Table, Node 26

|                            |                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2233-19                                                                                                                                                                                |
| Customer Symptom           | ORVM Co-driver horizontal axis motor movement stops                                                                                                                                     |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                                                                                                                                    |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                     |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                                                                                                                                    |
| Normal Operating Condition | ORVM Co-driver mirror works fine                                                                                                                                                        |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. ORVM Co-Driver horizontal axis motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

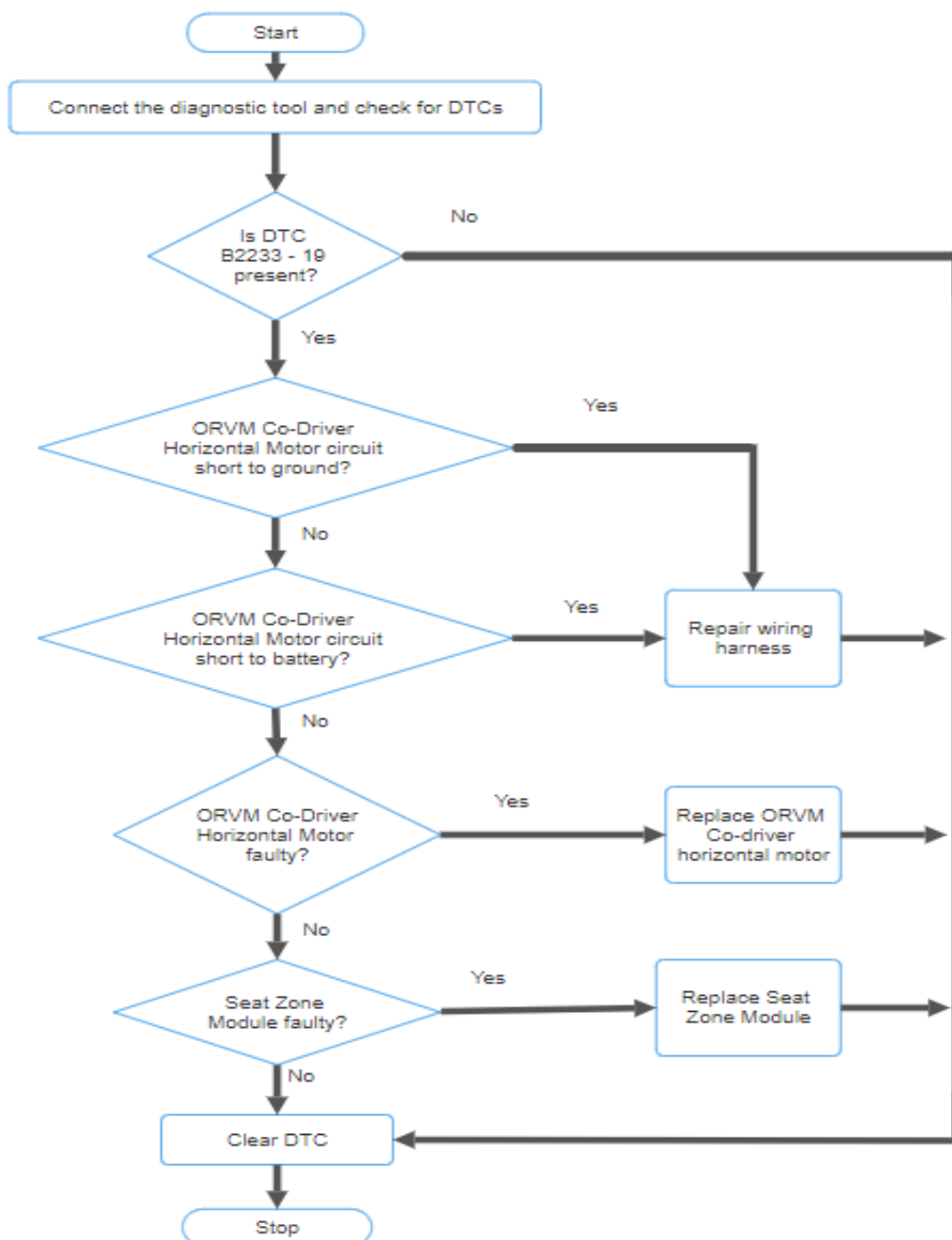
Version: 1.4

Date: 03-05-24

Text, Node 26

**Trouble Shooting: -**

Image, Node 26



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 26

#### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

#### **Circuit Schematic Diagram: -**

N.A

#### **Component Location & Image: -**

N.A

#### **Connector Location: -**

N.A

#### **Connector View & Information: -**

N.A

#### **Inspection Method: -**

N.A



DTC\_Code, Node 27

**B2233-01:- ORVM Co-Driver Horizontal axis Motor- general feedback failure**

Text, Node 27

**Overview:**

Table, Node 27

|                            |                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2233-01                                                                                                                                                                                       |
| Customer Symptom           | ORVM Co-driver horizontal axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                                                                                                                                           |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                            |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                                                                                                                                           |
| Normal Operating Condition | ORVM Co-driver mirror works fine                                                                                                                                                               |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Co-Driver Horizontal axis</u> motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                       |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

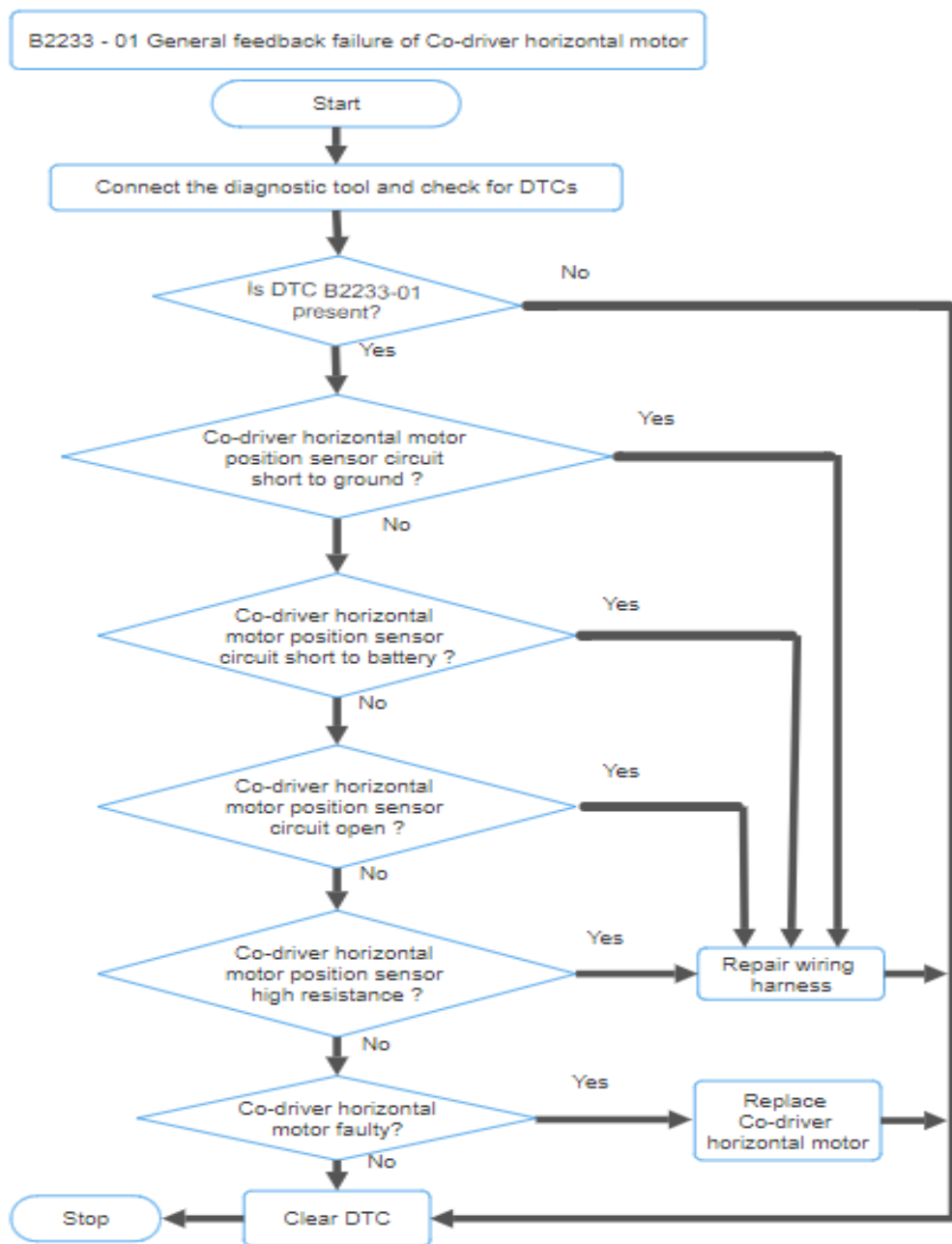
Version: 1.4

Date: 03-05-24

Text, Node 27

**Trouble Shooting: -**

Image, Node 27



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 27

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A



DTC\_Code, Node 27

**B2233-18:- ORVM Co-Driver horizontal axis motor under current**

Text, Node 27

**Overview:**

Table, Node 27

|                            |                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2233-18                                                                                                                                                                                       |
| Customer Symptom           | ORVM Co-driver horizontal axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                                                                                                                                           |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                            |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                                                                                                                                           |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                                                                                                                                                      |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Co-Driver horizontal axis motor</u> internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                       |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

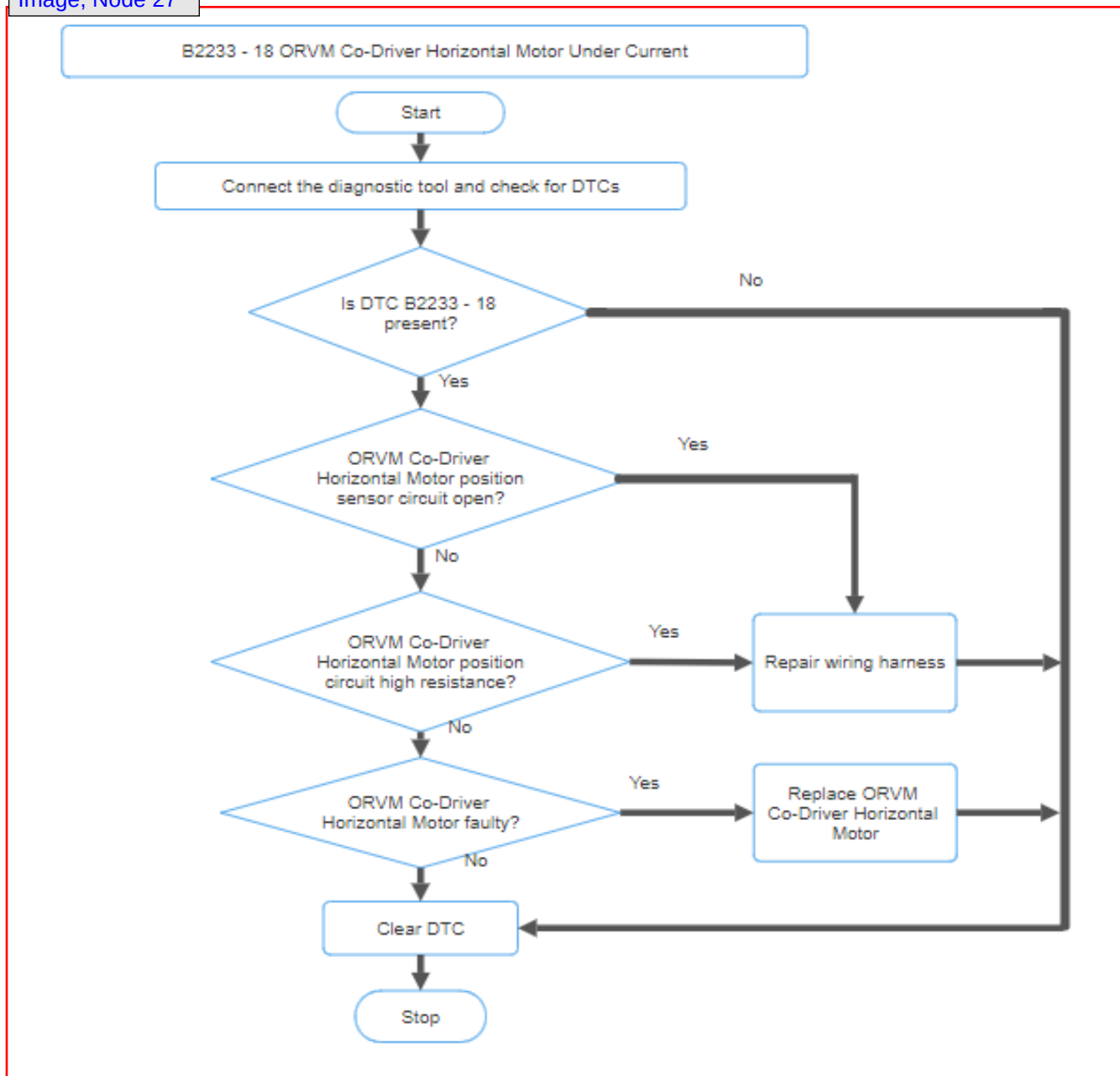
Version: 1.4

Date: 03-05-24

Text, Node 27

**Trouble Shooting: -**

Image, Node 27







**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 27

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



DTC\_Code, Node 28

**B2231-13:- ORVM Co-Driver vertical axis motor Open Circuit**

Text, Node 28

**Overview:**

Table, Node 28

|                            |                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2231-13                                                                                                                                                                                     |
| Customer Symptom           | ORVM Co-driver vertical axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                          |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                                                                                                                                         |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                                                                                                                                                    |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Co-Driver vertical axis motor internal failure</u> |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                     |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

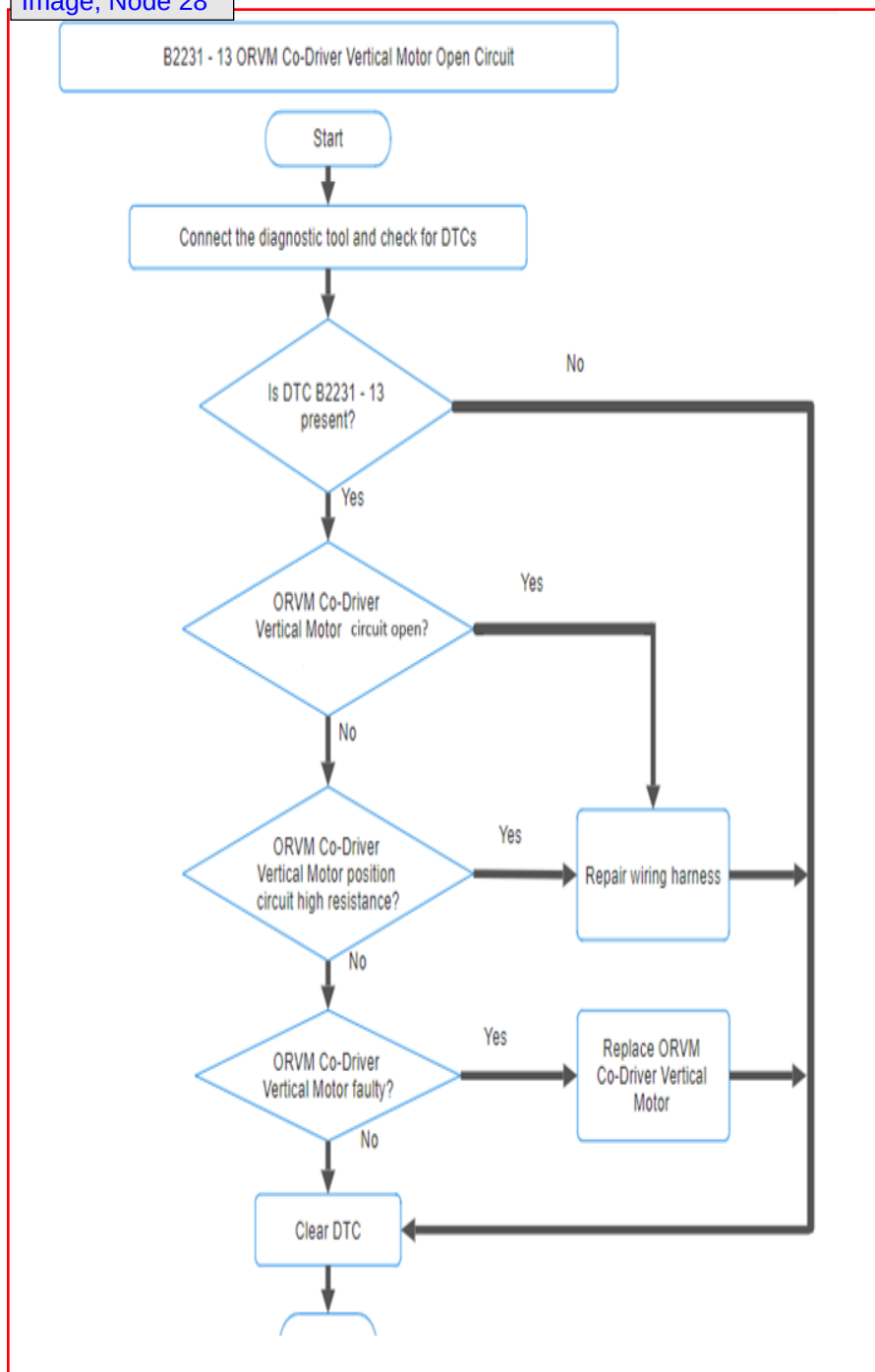
Version: 1.4

Date: 03-05-24

Text, Node 28

**Trouble Shooting: -**

Image, Node 28



Author: BCM COC

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**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 28

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 29

**B2231-11: ORVM Co-Driver vertical axis motor STG**

Text, Node 29

### Overview:

Table, Node 29

|                            |                                                                      |
|----------------------------|----------------------------------------------------------------------|
| ISO Code                   | B2231-11                                                             |
| Customer Symptom           | ORVM Co-driver vertical axis motor movement stops                    |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                 |
| Lamp Status (If Any)       | N.A                                                                  |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                 |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                            |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                             |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

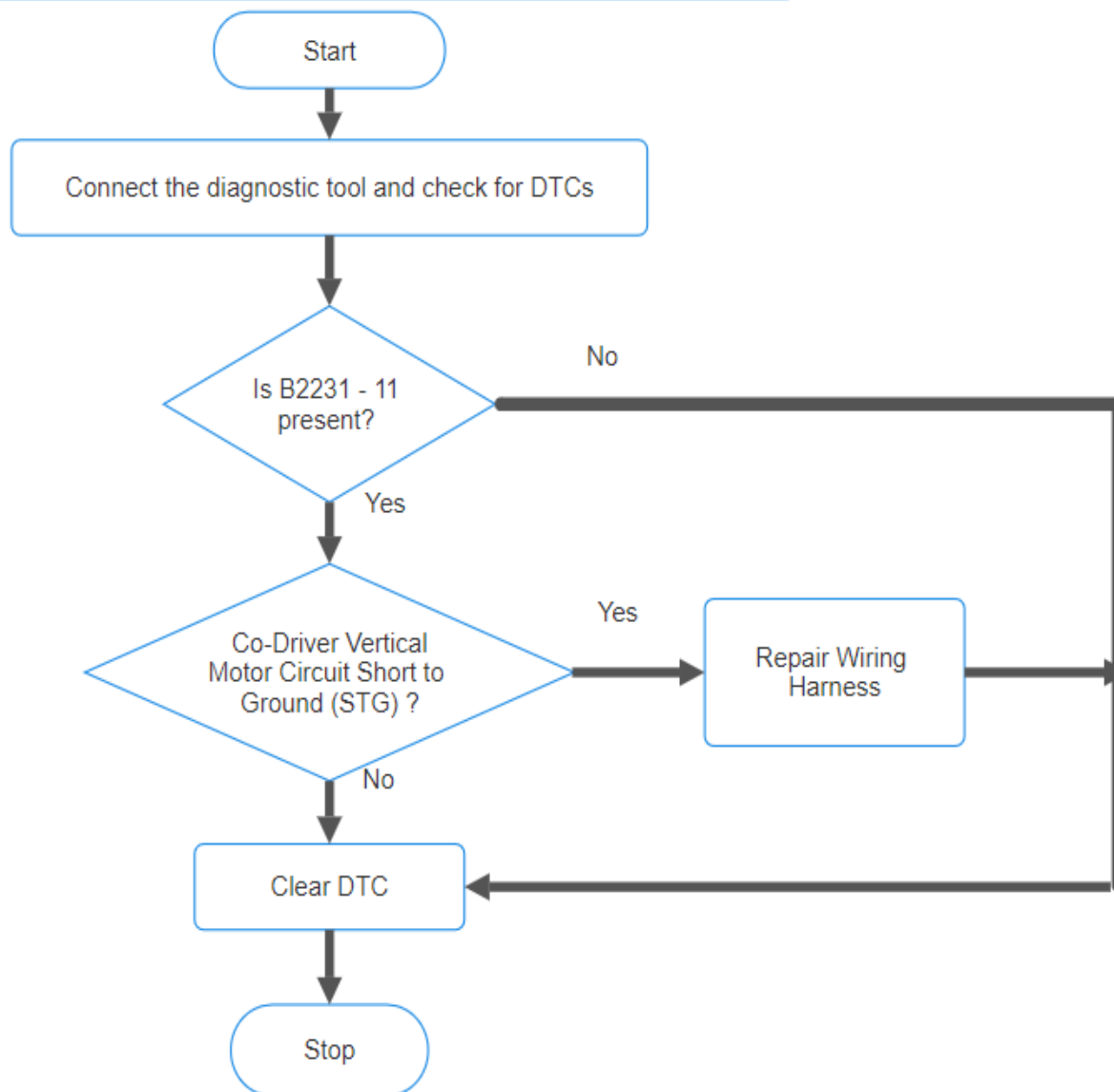
Date: 03-05-24

Text, Node 29

**Trouble Shooting: -**

Image, Node 29

B2231-11 ORVM Co-Driver Vertical Motor Circuit Short to Ground (STG)





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 29

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 30

### **B2231-12:- ORVM Co-Driver vertical axis motor STB**

Text, Node 30

#### **Overview:**

Table, Node 30

|                            |                                                                       |
|----------------------------|-----------------------------------------------------------------------|
| ISO Code                   | B2231-12                                                              |
| Customer Symptom           | ORVM Co-driver vertical axis motor movement stops                     |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops.                                 |
| Lamp Status (If Any)       | N.A                                                                   |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                  |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                             |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to Battery |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                              |





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

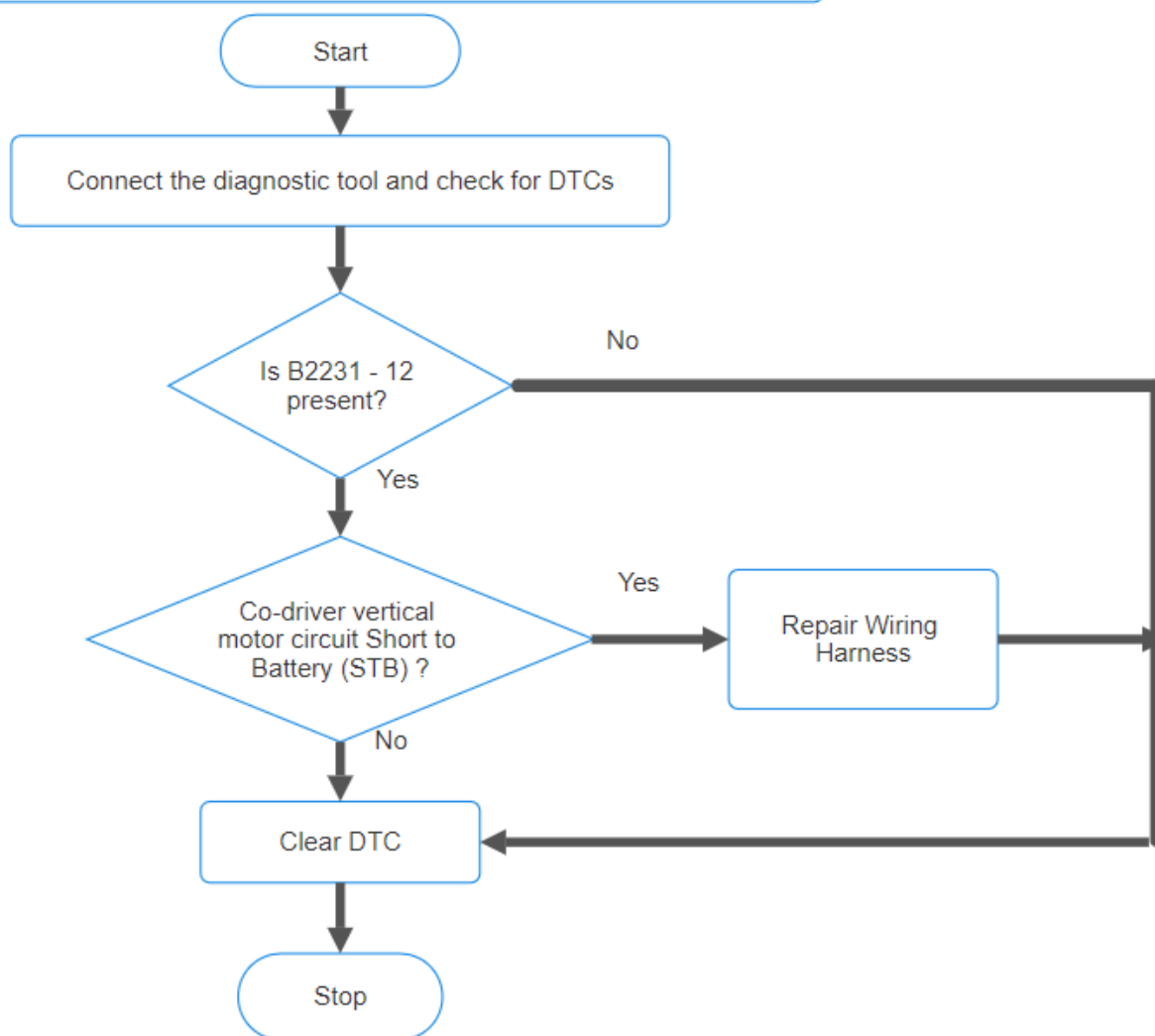
Date: 03-05-24

Text, Node 30

**Trouble Shooting: -**

Image, Node 30

B2231 - 12 ORVM Co-Driver Vertical Motor Circuit Short to Battery (STB)



Author: BCM COC

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|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 30

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A



|                 |                 |
|-----------------|-----------------|
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|-----------------|-----------------|

DTC\_Code, Node 31

**B2231-19:- ORVM Co-Driver vertical axis motor Over current**

Text, Node 31

**Overview:**

Table, Node 31

|                            |                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2231-19                                                                                                                                                                                     |
| Customer Symptom           | ORVM Co-driver vertical axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                          |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                                                                                                                                         |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                                                                                                                                                    |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Co-Driver vertical axis motor internal failure</u> |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                     |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

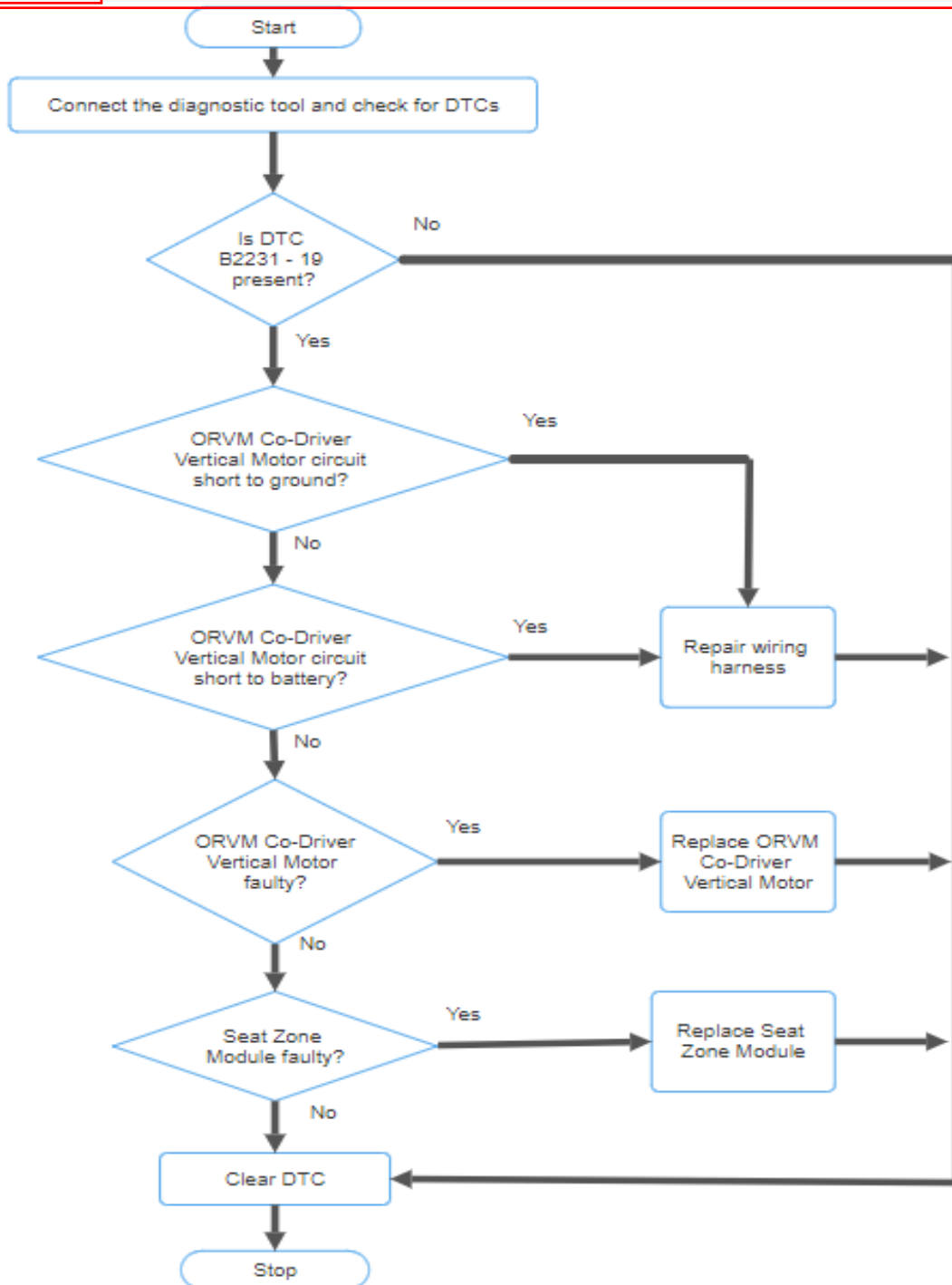
Version: 1.4

Date: 03-05-24

Text, Node 31

**Trouble Shooting: -**

Image, Node 31



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 31

### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### **Circuit Schematic Diagram: -**

N.A

### **Component Location & Image: -**

N.A

### **Connector Location: -**

N.A

### **Connector View & Information: -**

N.A

### **Inspection Method: -**

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 32

**B2231-01:- ORVM Co-Driver Vertical Axis Motor- General Feedback Failure**

Table, Node 32

|                            |                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2231-01                                                                                                                                                                                     |
| Customer Symptom           | ORVM co-driver vertical axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM co-driver mirror Movement Stops                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                          |
| Fault Detection Condition  | ORVM co-driver mirror Movement Stops                                                                                                                                                         |
| Normal Operating Condition | ORVM co-driver mirror Movement works fine                                                                                                                                                    |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Co-Driver Vertical Axis</u> motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                     |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

Version: 1.4

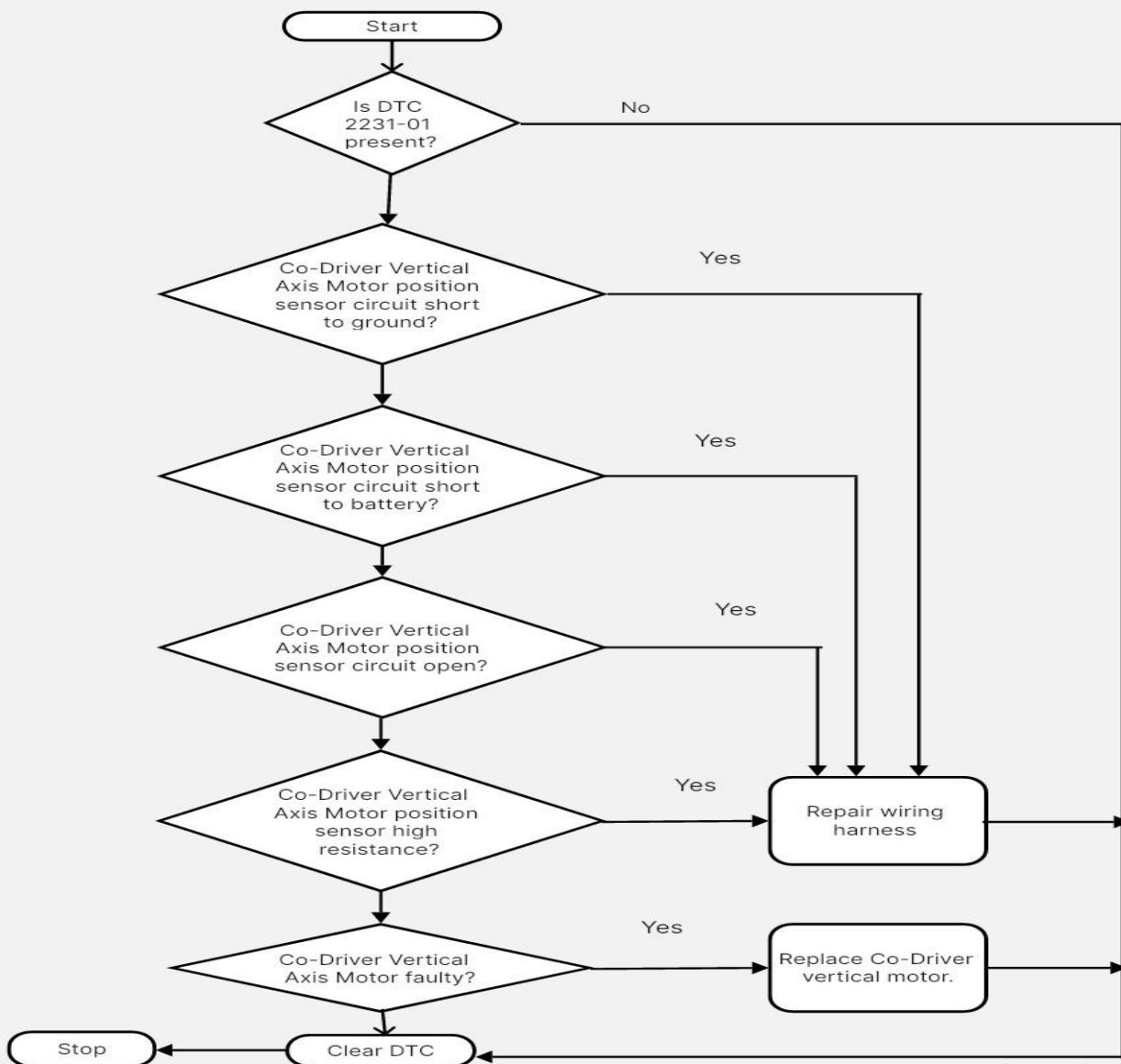
Date: 03-05-24

Text, Node 32

**Trouble Shooting: -**

Image, Node 32

B 2231-01 ORVM Co-Driver Vertical Axis Motor - General Feedback Failure



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|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 32

### DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### Circuit Schematic Diagram: -

N.A

### Component Location & Image: -

N.A

### Connector Location: -

N.A

### Connector View & Information: -

N.A

### Inspection Method: -

N.A







**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 33

**B2231-18:- ORVM Co-Driver vertical axis motor under current**

Text, Node 33

**Overview:**

Table, Node 33

|                            |                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2231-18                                                                                                                                                                           |
| Customer Symptom           | ORVM Co-driver vertical axis motor movement stops                                                                                                                                  |
| Fault Effects (On Vehicle) | ORVM Co-driver mirror Movement Stops                                                                                                                                               |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                |
| Fault Detection Condition  | ORVM Co-driver mirror Movement Stops                                                                                                                                               |
| Normal Operating Condition | ORVM Co-driver mirror Movement works fine                                                                                                                                          |
| Probable Trouble Area      | ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. ORVM Co-Driver vertical axis motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                           |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

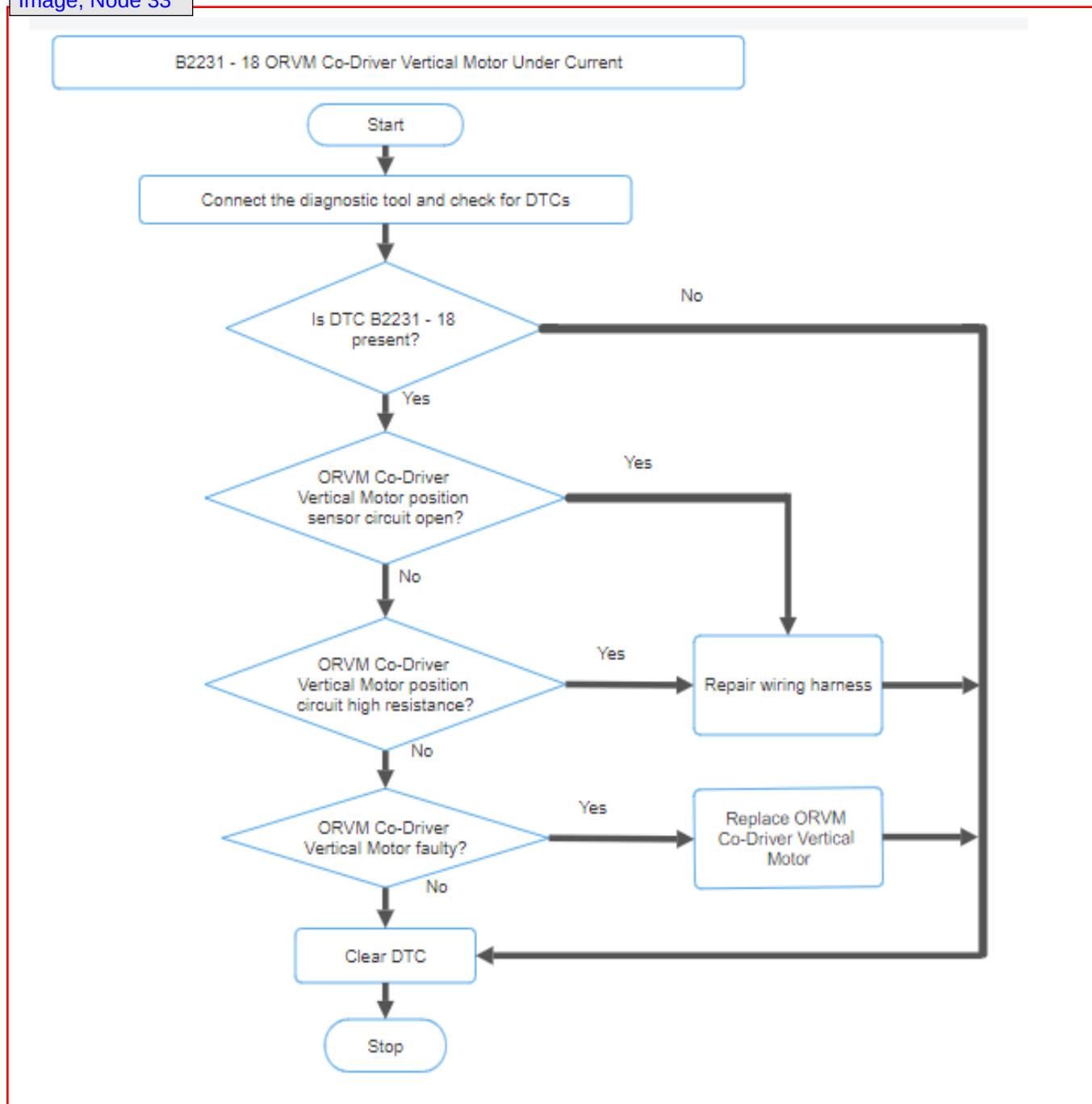
Version: 1.4

Date: 03-05-24

Text, Node 33

**Trouble Shooting: -**

Image, Node 33





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 33

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



DTC\_Code, Node 34

**B2207-13:- ORVM Driver horizontal axis motor Open Circuit**

Text, Node 34

**Overview:**

Table, Node 34

|                            |                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2207-13                                                                                                                                                                             |
| Customer Symptom           | ORVM driver horizontal axis motor movement stops                                                                                                                                     |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                    |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                  |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                    |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                               |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. ORVM Driver horizontal axis motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                             |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

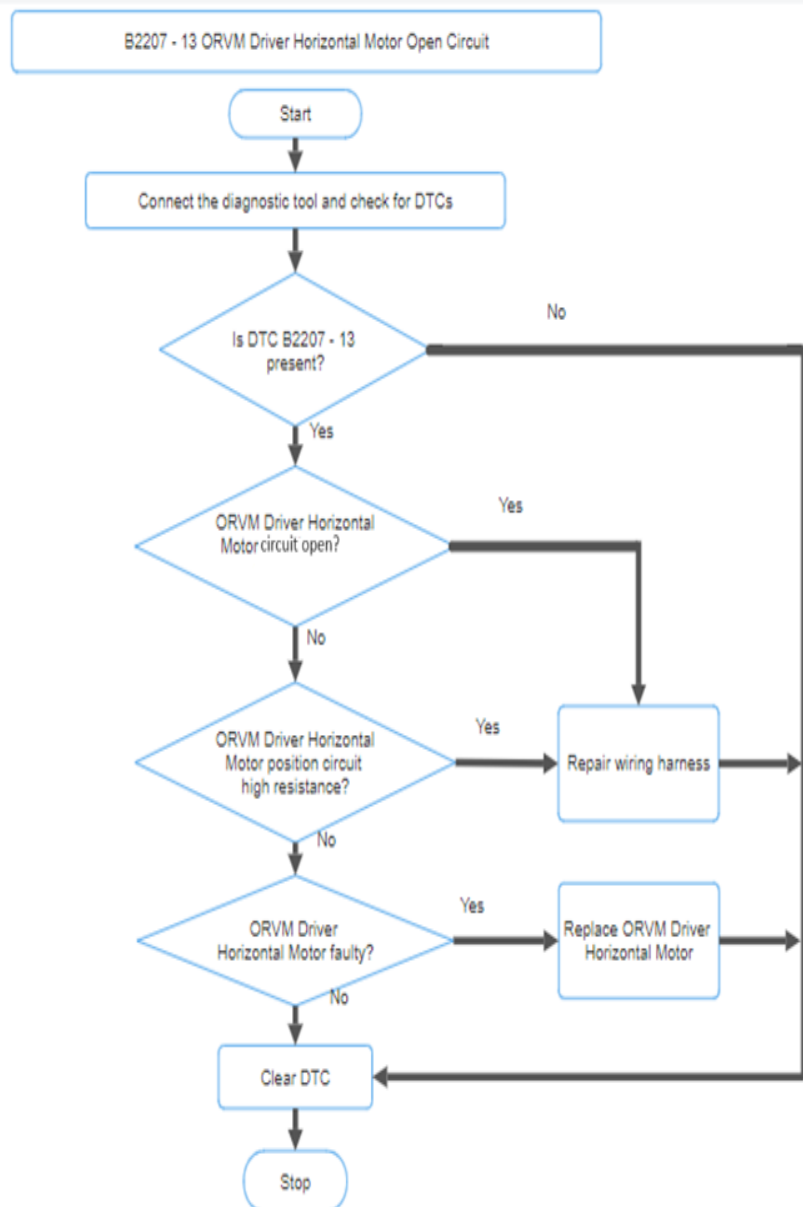
Version: 1.4

Date: 03-05-24

Text, Node 34

**Trouble Shooting: -**

Image, Node 34



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 34

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 35

### **B2207-11:- ORVM Driver horizontal axis motor STG**

Text, Node 35

#### **Overview:**

Table, Node 35

|                            |                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------|
| ISO Code                   | B2207-11                                                                                         |
| Customer Symptom           | ORVM driver horizontal axis motor movement stops                                                 |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                |
| Lamp Status (If Any)       | N.A                                                                                              |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                           |
| Probable Trouble Area      | ORVM mirror motor position sensor circuit short circuit to ground                                |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair wiring harness</li> <li>2. Clear DTC</li> </ol> |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

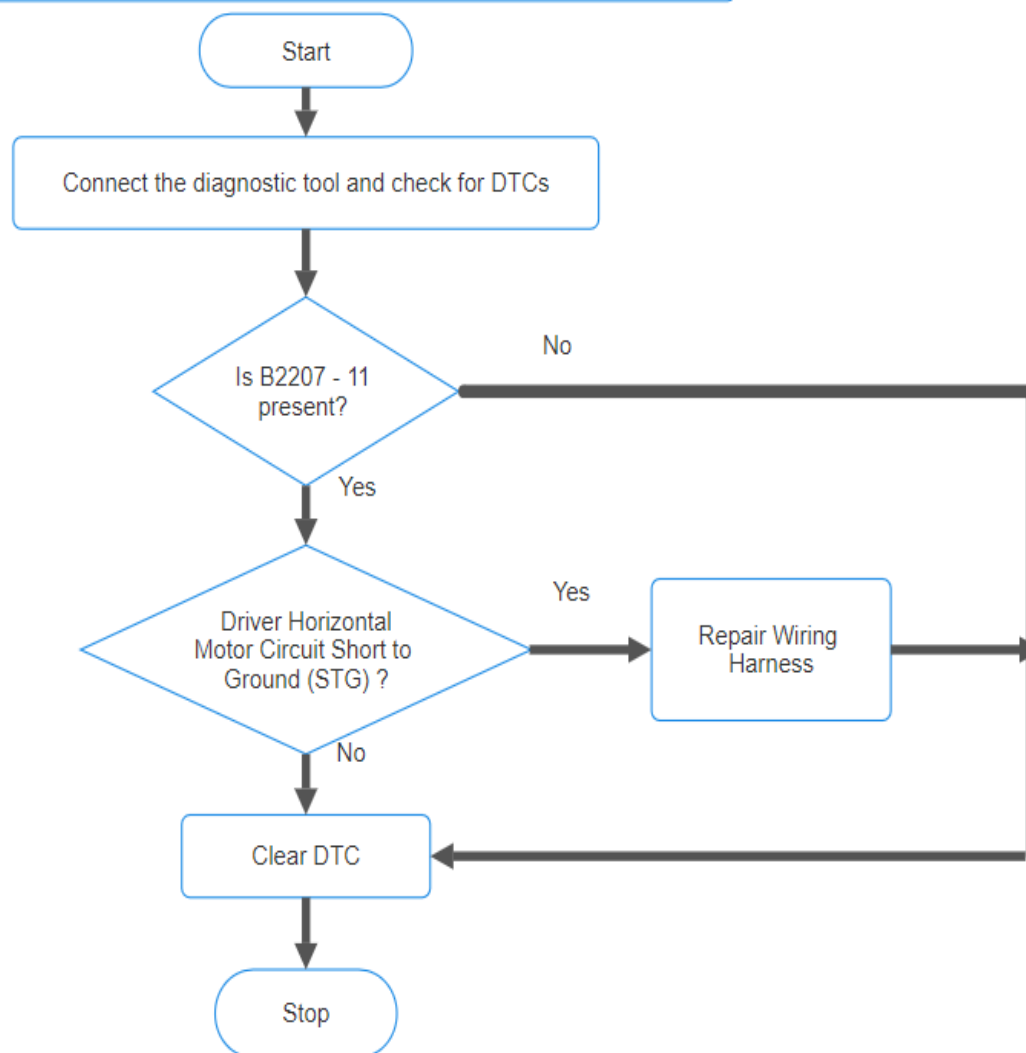
Date: 03-05-24

Text, Node 35

**Trouble Shooting: -**

Image, Node 35

B2207 -11 ORVM Driver Horizontal Motor Circuit Short to Ground (STG)



Author: BCM COC

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|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 35

### DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### Circuit Schematic Diagram: -

N.A

### Component Location & Image: -

N.A

### Connector Location: -

N.A

### Connector View & Information: -

N.A

### Inspection Method: -

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 36

**B2207-12:- ORVM Driver horizontal axis motor STB**

Text, Node 36

### Overview:

Table, Node 36

|                            |                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------|
| ISO Code                   | B2207-12                                                                                         |
| Customer Symptom           | ORVM driver horizontal axis motor movement stops                                                 |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                |
| Lamp Status (If Any)       | N.A                                                                                              |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                           |
| Probable Trouble Area      | 1.Seat forward/reward motor position sensor circuit short circuit to Battery                     |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Repair wiring harness</li> <li>2. Clear DTC</li> </ol> |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

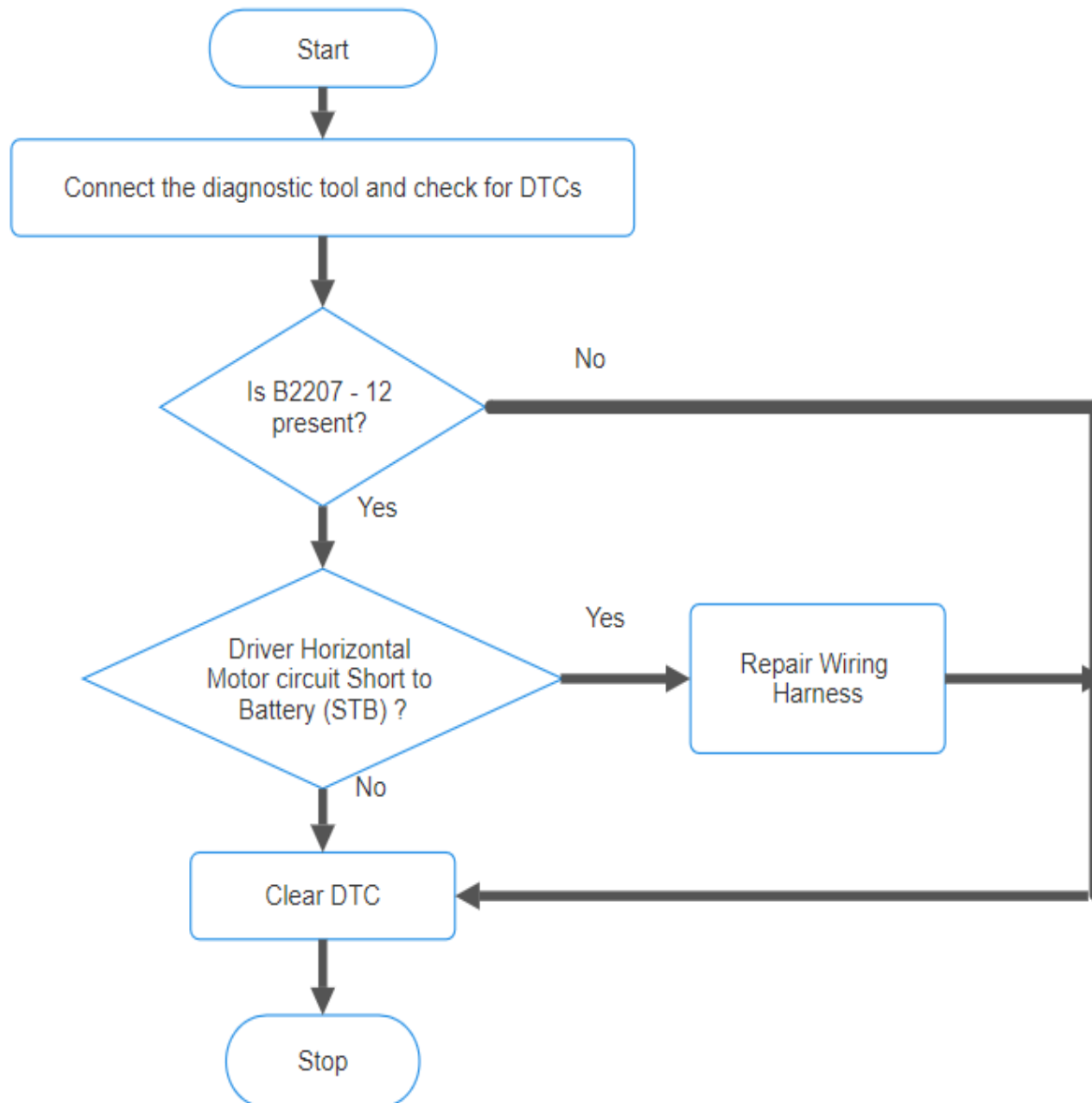
Date: 03-05-24

Text, Node 36

**Trouble Shooting: -**

Image, Node 36

B2207 - 12 ORVM Driver Horizontal Motor Circuit Short to Battery (STB)



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 36

### DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### Circuit Schematic Diagram: -

N.A

### Component Location & Image: -

N.A

### Connector Location: -

N.A

### Connector View & Information: -

N.A

### Inspection Method: -

N.A



DTC\_Code, Node 37

**B2207-19:- ORVM Driver horizontal axis motor Over current**

Text, Node 37

**Overview:**

Table, Node 37

|                            |                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2207-19                                                                                                                                                                             |
| Customer Symptom           | ORVM driver horizontal axis motor movement stops                                                                                                                                     |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                    |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                  |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                    |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                               |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. ORVM Driver horizontal axis motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                             |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

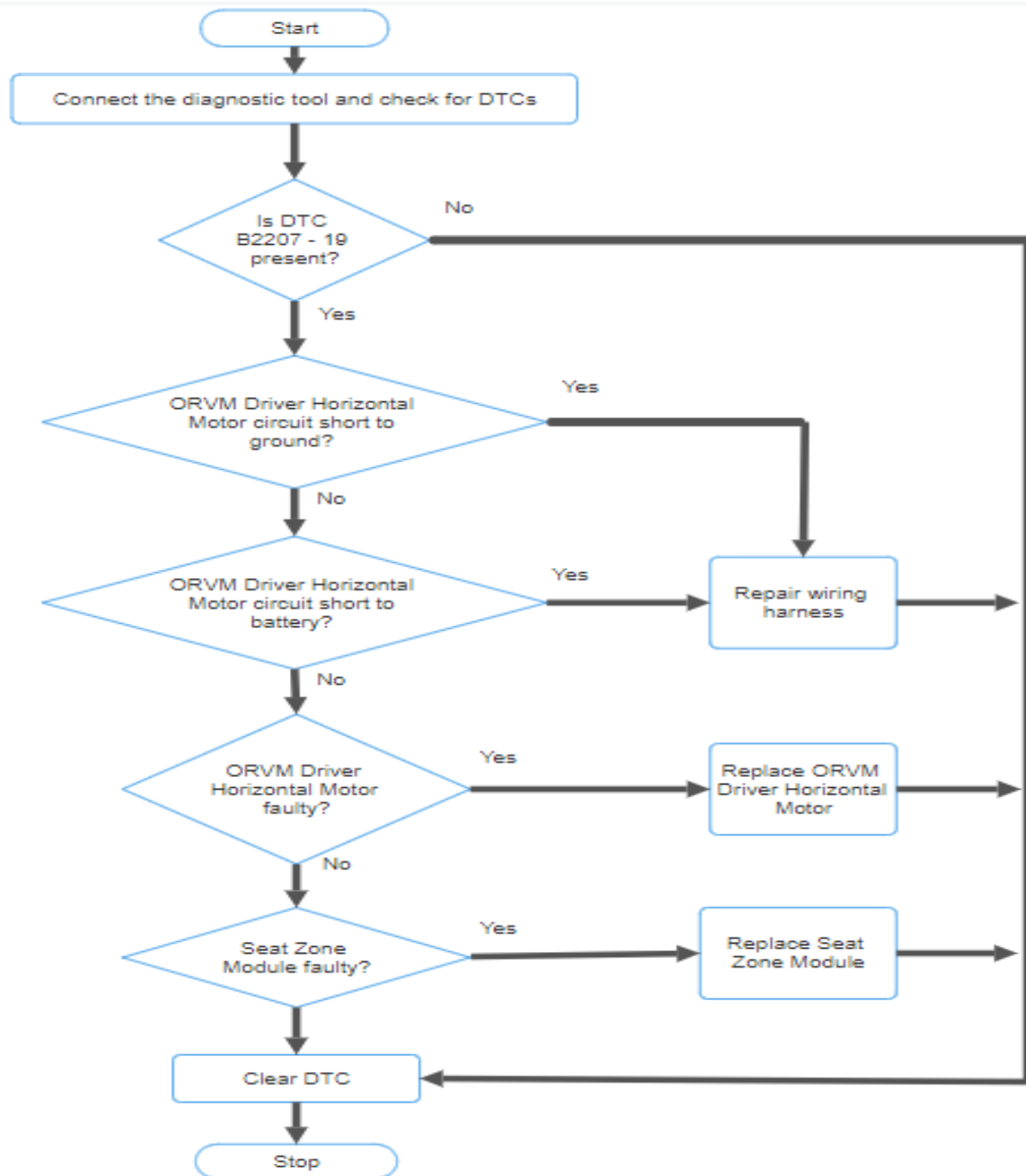
Version: 1.4

Date: 03-05-24

Text, Node 37

**Trouble Shooting: -**

Image, Node 37



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 37

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



DTC\_Code, Node 38

**B2207-01:- Driver horizontal axis motor**

Text, Node 38

**Overview:**

Table, Node 38

|                            |                                                                                                                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2207-01                                                                                                                                                                               |
| Customer Symptom           | ORVM driver horizontal axis motor movement stops                                                                                                                                       |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                      |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                    |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                      |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                                 |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>driver horizontal axis motor</u> internal failure |
| Healing Condition          | 3. Repair wiring harness<br>4. Clear DTC                                                                                                                                               |





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

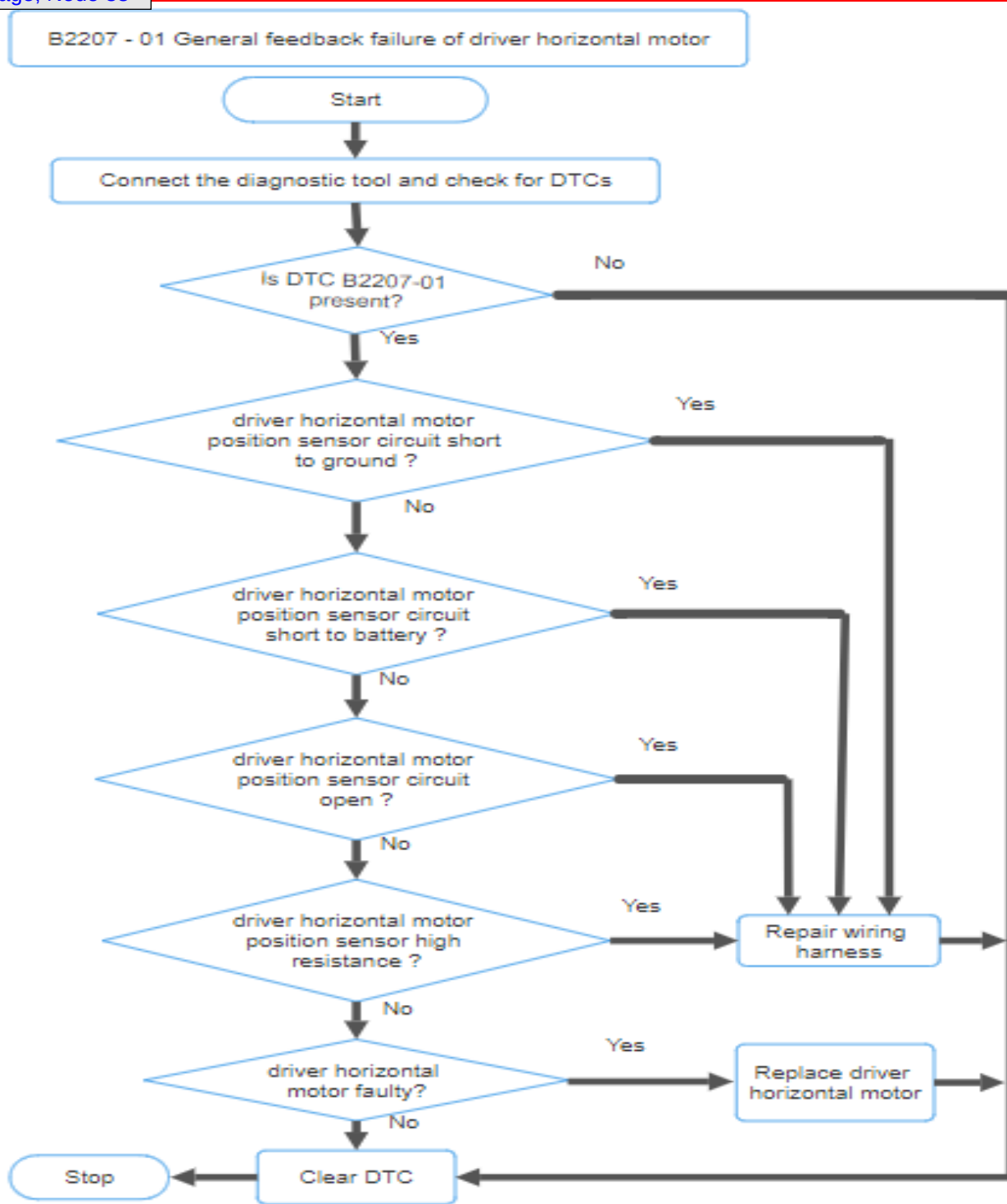
Version: 1.4

Date: 03-05-24

Text, Node 38

**Trouble Shooting: -**

Image, Node 38



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 38

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 39

### **B2207-18:- ORVM Driver horizontal axis motor under current**

Text, Node 39

#### **Overview:**

Table, Node 39

|                            |                                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2207-18                                                                                                                                                                                    |
| Customer Symptom           | ORVM driver horizontal axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                           |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                         |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                           |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                                      |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Driver horizontal axis motor</u> internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                    |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

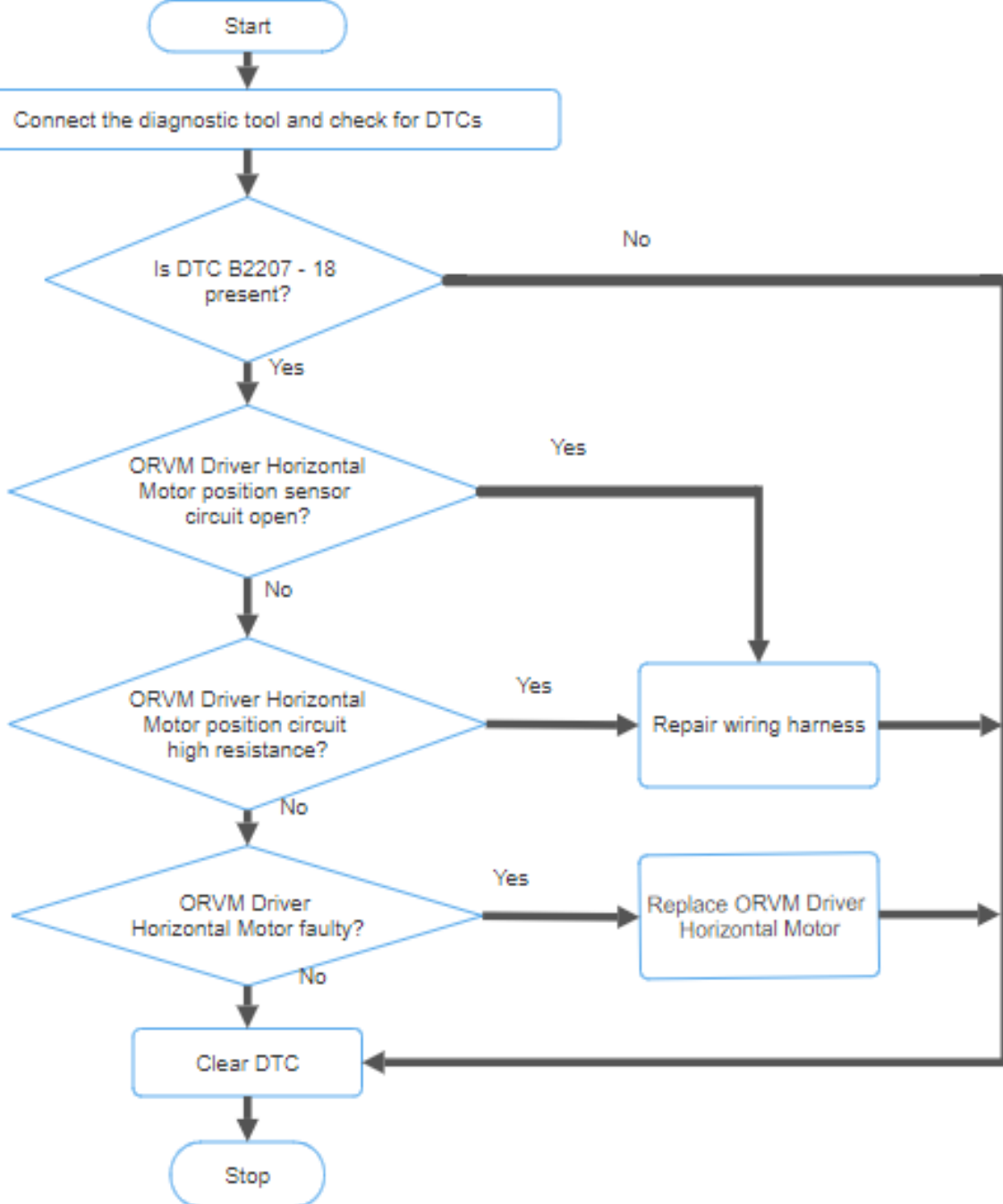
Date: 03-05-24

Text, Node 39

**Trouble Shooting: -**

Image, Node 39

B2207 - 18 ORVM Driver Horizontal Motor Under Current



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 39

#### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

#### **Circuit Schematic Diagram: -**

N.A

#### **Component Location & Image: -**

N.A

#### **Connector Location: -**

N.A

#### **Connector View & Information: -**

N.A

#### **Inspection Method: -**

N.A



DTC\_Code, Node 40

**B2203-13:- ORVM Driver vertical axis motor Open Circuit**

Text, Node 40

**Overview:**

Table, Node 40

|                            |                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2203-13                                                                                                                                                                           |
| Customer Symptom           | ORVM driver vertical axis motor movement stops                                                                                                                                     |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                  |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                  |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                             |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. ORVM Driver vertical axis motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                           |



# Vehicle Platform: ECU Name: Seat Zone Module (SZM) DTC Troubleshooting Data

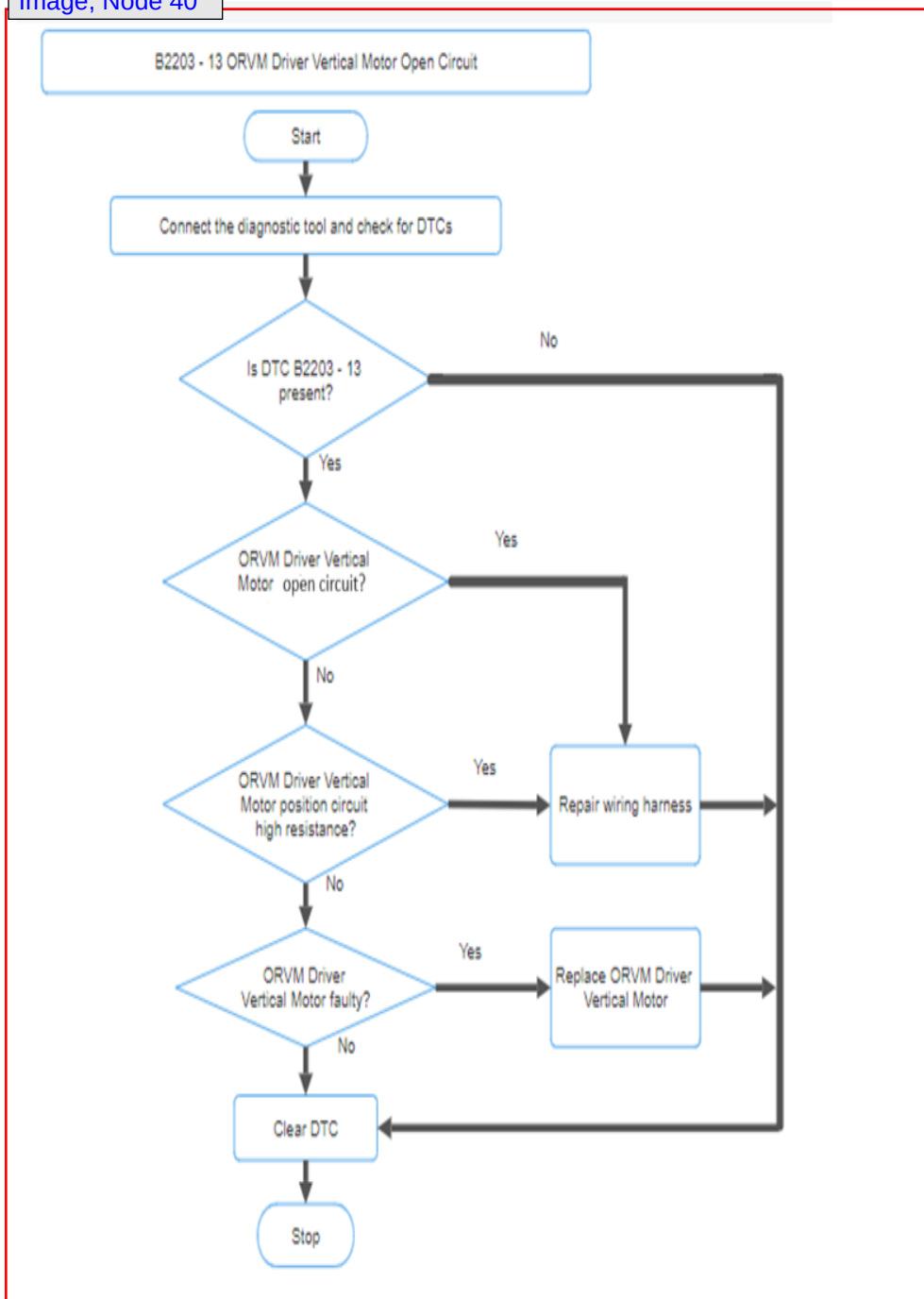
Version: 1.4

Date: 03-05-24

Text, Node 40

**Trouble Shooting: -**

Image, Node 40





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 40

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A





DTC\_Code, Node 41

**B1F6B-11:- SZM ORVM Driver vertical axis motor STC**

Text, Node 41

**Overview:**

Table, Node 41

|                            |                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------|
| ISO Code                   | B1F6B-11                                                                               |
| Customer Symptom           | ORVM driver vertical axis motor movement stops                                         |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                      |
| Lamp Status (If Any)       | N.A                                                                                    |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                      |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                 |
| Probable Trouble Area      | 1. SZM ORVM Driver vertical axis_motor position sensor circuit short circuit to ground |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                               |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

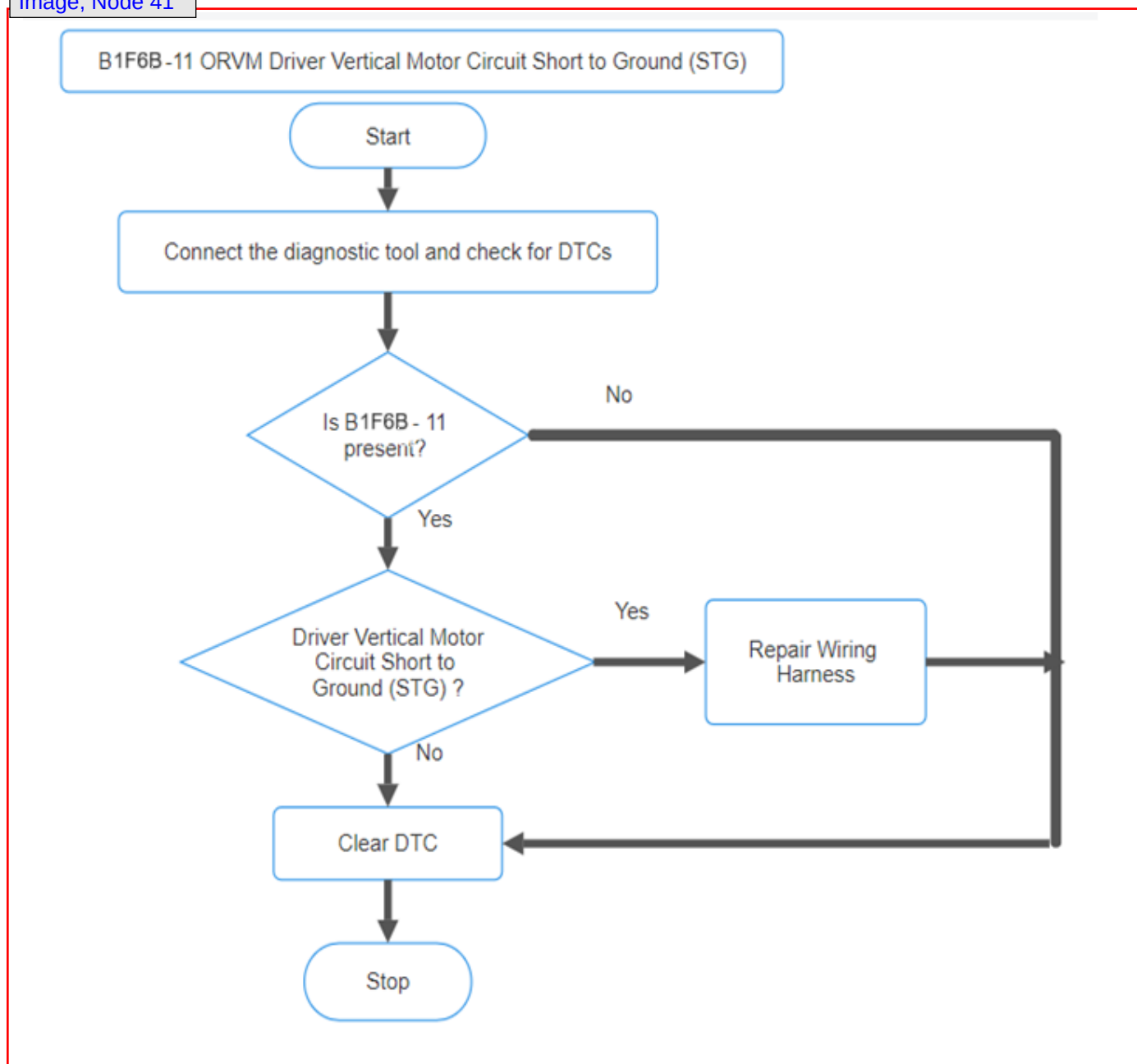
Version: 1.4

Date: 03-05-24

Text, Node 41

**Trouble Shooting: -**

Image, Node 41



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 41

### DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### Circuit Schematic Diagram: -

N.A

### Component Location & Image: -

N.A

### Connector Location: -

N.A

### Connector View & Information: -

N.A

### Inspection Method: -

N.A



DTC\_Code, Node 42

**B2203-15:- ORVM Driver vertical axis motor STB**

Text, Node 42

**Overview:**

Table, Node 42

|                            |                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------|
| ISO Code                   | B2203-15                                                                            |
| Customer Symptom           | ORVM driver vertical axis motor movement stops                                      |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                   |
| Lamp Status (If Any)       | N.A                                                                                 |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                   |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                              |
| Probable Trouble Area      | 1. ORVM Driver vertical axis motor position sensor circuit short circuit to Battery |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                            |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

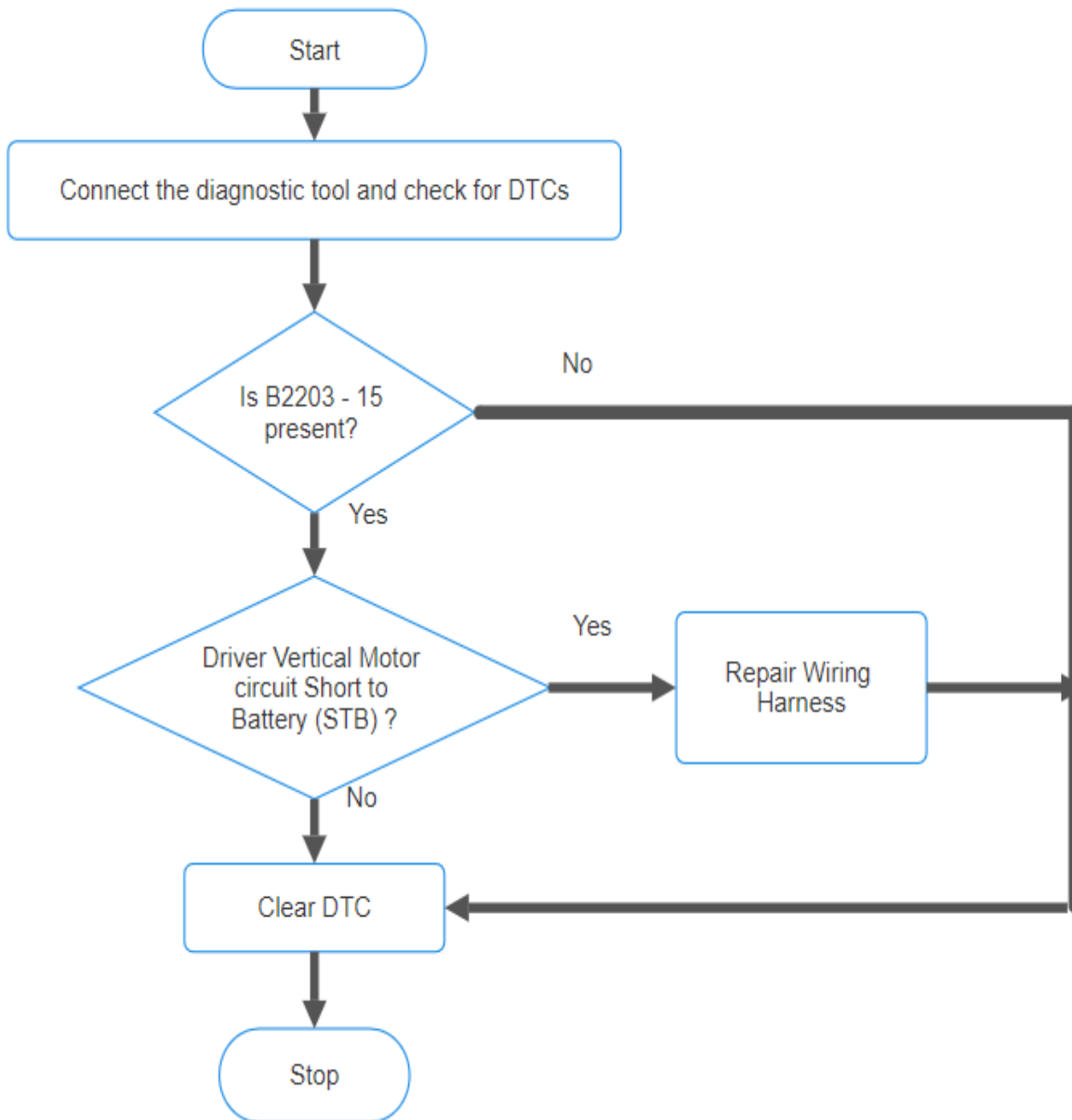
Date: 03-05-24

Text, Node 42

**Trouble Shooting: -**

Image, Node 42

B2203 - 15 ORVM Driver Vertical Motor Circuit Short to Battery (STB)



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 42

### DTC's confirmation: -

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

### Circuit Schematic Diagram: -

N.A

### Component Location & Image: -

N.A

### Connector Location: -

N.A

### Connector View & Information: -

N.A

### Inspection Method: -

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 43

**B2203-19: ORVM Driver vertical axis motor Over current**

Text, Node 43

### Overview:

Table, Node 43

|                            |                                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2203-19                                                                                                                                                                                  |
| Customer Symptom           | ORVM driver vertical axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                       |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                         |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                                    |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Driver vertical axis motor</u> internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                  |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

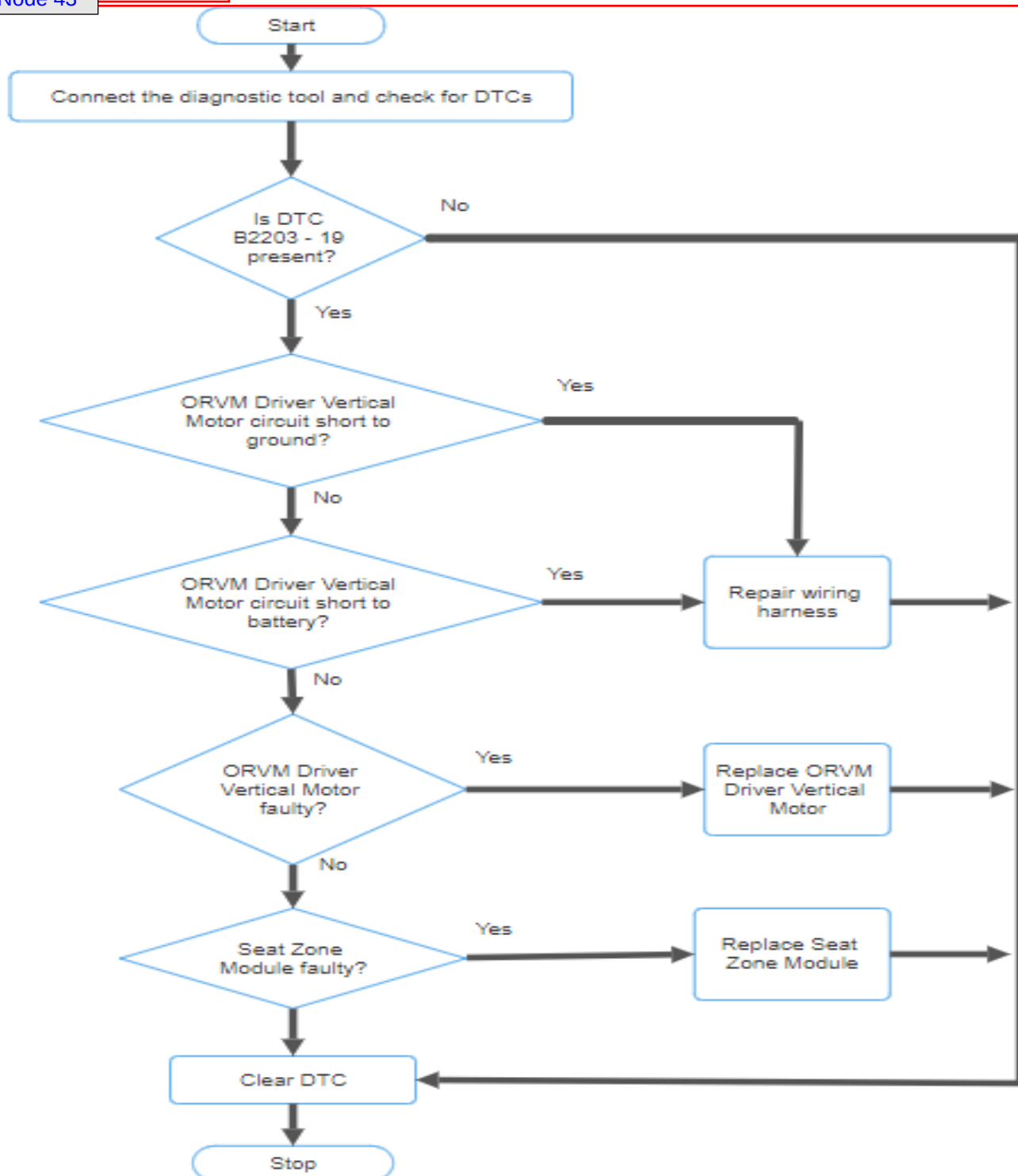
Version: 1.4

Date: 03-05-24

Text, Node 43

**Trouble Shooting: -**

Image, Node 43





|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 43

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 44

**B2203-01:- Driver vertical axis motor- General Feedback Failure**

Text, Node 44

### Overview:

Table, Node 44

|                            |                                                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B155A-12                                                                                                                                                                      |
| Customer Symptom           | ORVM driver vertical axis motor movement stops                                                                                                                                |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                             |
| Lamp Status (If Any)       | N.A                                                                                                                                                                           |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                             |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                        |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. driver vertical axis motor internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                      |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

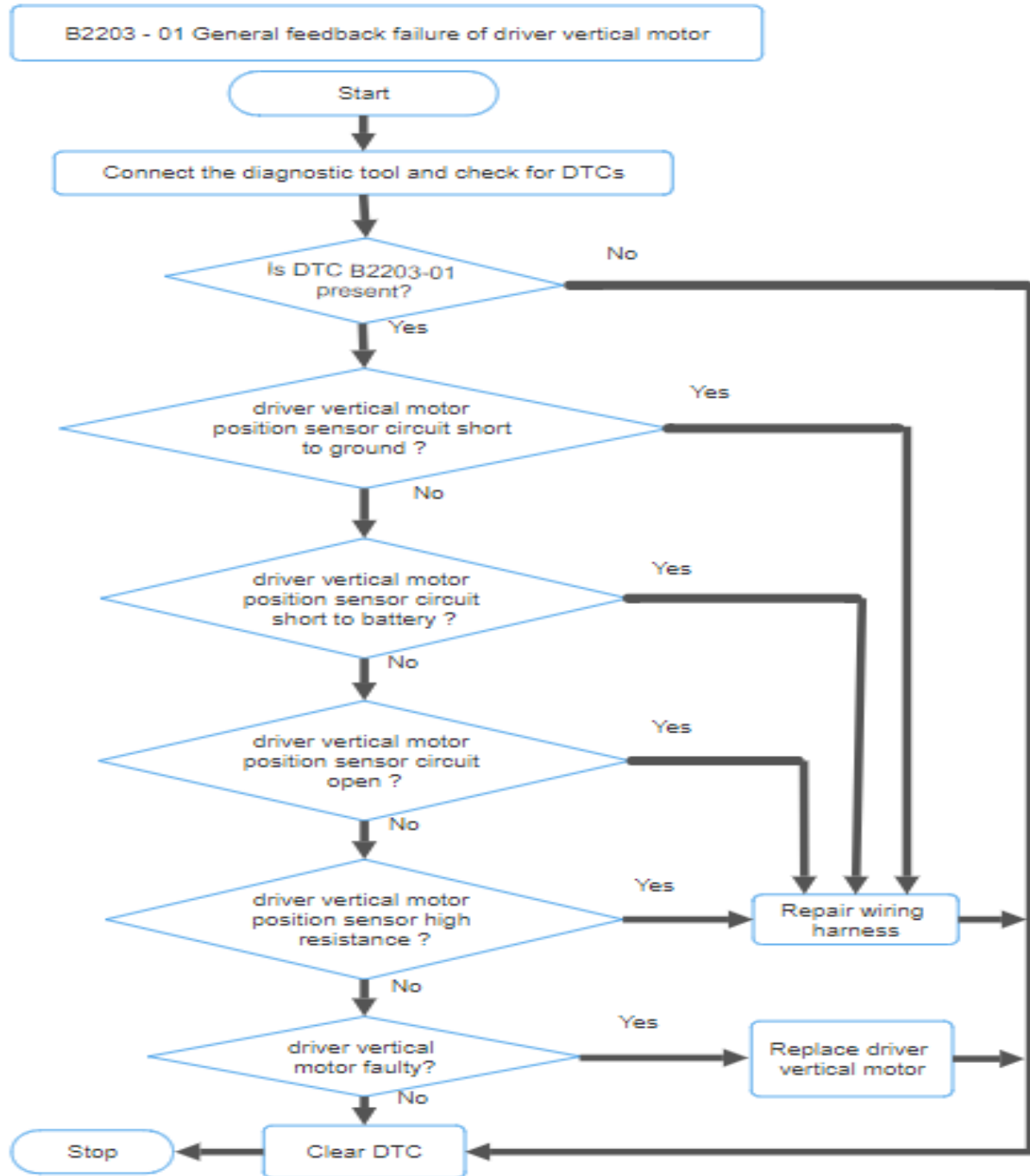
Version: 1.4

Date: 03-05-24

Text, Node 44

**Trouble Shooting: -**

Image, Node 44



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 44

#### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

#### **Circuit Schematic Diagram: -**

N.A

#### **Component Location & Image: -**

N.A

#### **Connector Location: -**

N.A

#### **Connector View & Information: -**

N.A

#### **Inspection Method: -**

N.A





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

DTC\_Code, Node 45

**B2203-18:- ORVM Driver vertical axis motor under current**

Text, Node 45

**Overview:**

Table, Node 45

|                            |                                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B2203-18                                                                                                                                                                                  |
| Customer Symptom           | ORVM driver vertical axis motor movement stops                                                                                                                                            |
| Fault Effects (On Vehicle) | ORVM driver mirror Movement Stops                                                                                                                                                         |
| Lamp Status (If Any)       | N.A                                                                                                                                                                                       |
| Fault Detection Condition  | ORVM driver mirror Movement Stops                                                                                                                                                         |
| Normal Operating Condition | ORVM driver mirror Movement works fine                                                                                                                                                    |
| Probable Trouble Area      | 1. ORVM mirror motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance<br>2. <u>ORVM Driver vertical axis motor</u> internal failure |
| Healing Condition          | 1. Repair wiring harness<br>2. Clear DTC                                                                                                                                                  |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

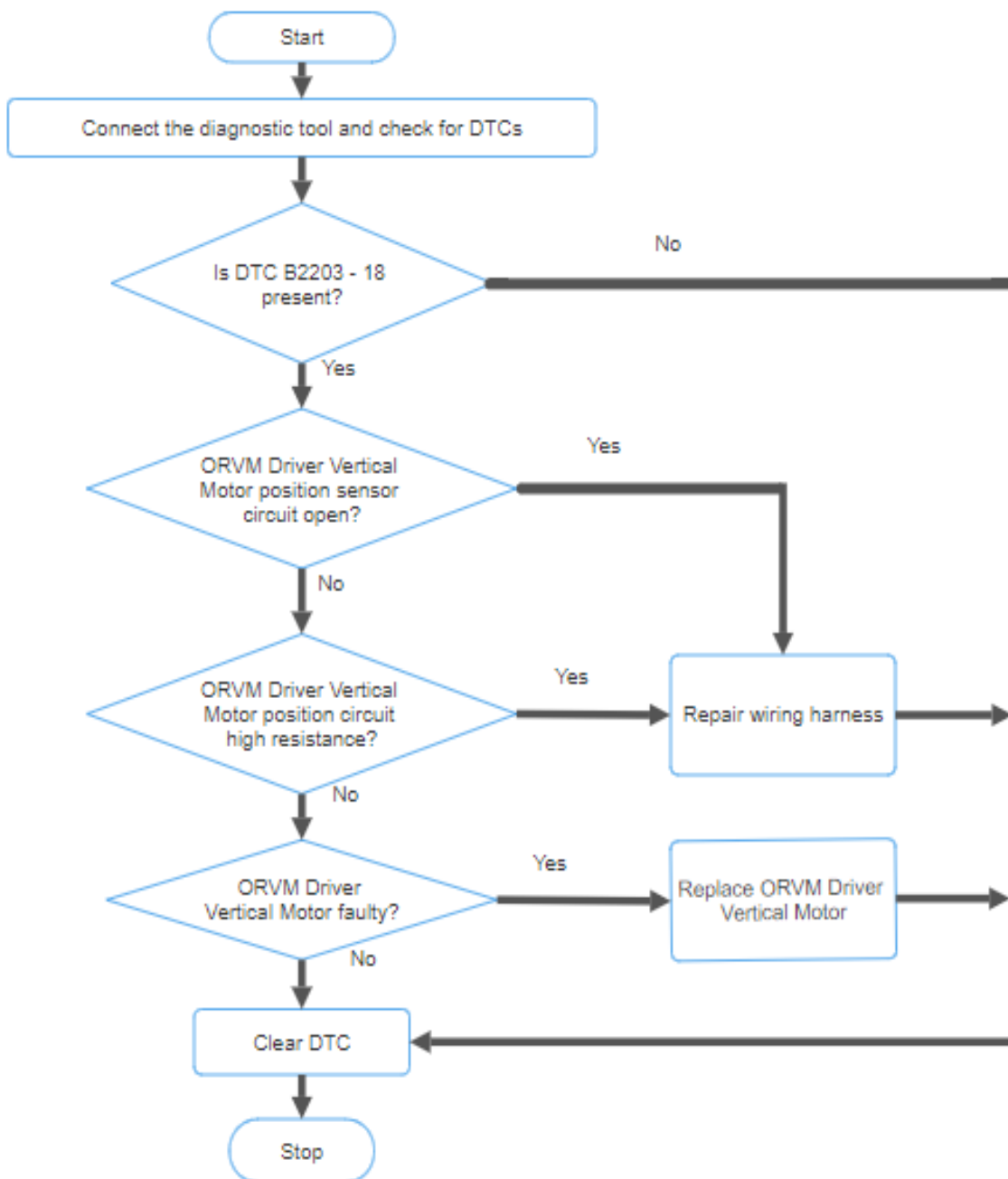
Date: 03-05-24

Text, Node 45

**Trouble Shooting: -**

Image, Node 45

B2203 - 18 ORVM Driver Vertical Motor Under Current





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

Date: 03-05-24

Text, Node 45

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A



DTC\_Code, Node 46

**B1F6A-71: ORVM Joystick Stuck- Actuator Stuck**

Text, Node 46

**Overview:**

Table, Node 46

|                            |                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1F6A-71                                                                                               |
| Customer Symptom           | ORVM Joystick stuck - Actuator stuck                                                                   |
| Fault Effects (On Vehicle) | ORVM Joystick stuck - Actuator stuck                                                                   |
| Lamp Status (If Any)       | N. A                                                                                                   |
| Fault Detection Condition  | ORVM Joystick Not Pressed/ Stuck                                                                       |
| Normal Operating Condition | ORVM Joystick getting Press/No stuck Condition                                                         |
| Probable Trouble Area      | 1. ORVM Joystick stuck active<br><br>2. ORVM Joystick internal failure                                 |
| Healing Condition          | 1. Free Stuck Switches<br>2. Install a new front seat switch pack as necessary<br>3. Clear DTC Command |





**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

Version: 1.4

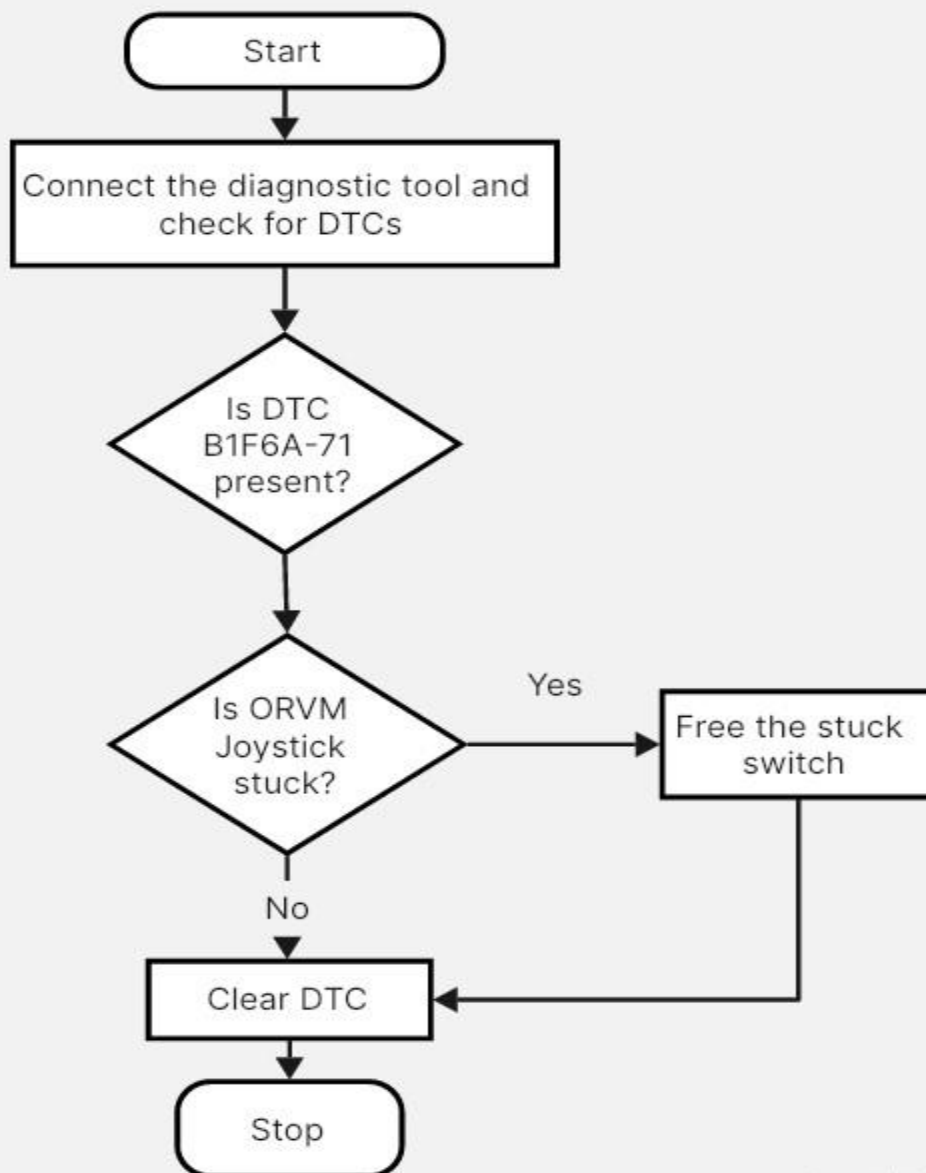
Date: 03-05-24

Text, Node 46

**Trouble Shooting: -**

Image, Node 46

B1F6A-71 ORVM Joystick stuck - Actuator stuck



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 46

#### **DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

#### **Circuit Schematic Diagram: -**

N.A

#### **Component Location & Image: -**

N.A

#### **Connector Location: -**

N.A

#### **Connector View & Information: -**

N.A

#### **Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

DTC\_Code, Node 47

### **B1F6A-13: ORVM Joystick Fault – Circuit Open**

Text, Node 47

#### **Overview:**

Table, Node 47

|                            |                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ISO Code                   | B1F6A-13                                                                                                                             |
| Customer Symptom           | When User presses joystick, ORVM does not move.                                                                                      |
| Fault Effects (On Vehicle) | ORVM does not move on pressing joystick.                                                                                             |
| Lamp Status (If Any)       | N. A                                                                                                                                 |
| Fault Detection Condition  | ORVM not moving because of joystick                                                                                                  |
| Normal Operating Condition | ORVM movement happens on operating joystick.                                                                                         |
| Probable Trouble Area      | <ol style="list-style-type: none"> <li>1. ORVM Joystick not connected to SZM</li> <li>2. ORVM Joystick input not received</li> </ol> |
| Healing Condition          | <ol style="list-style-type: none"> <li>1. Connect joystick to SZM</li> <li>2. Repair joystick.</li> </ol>                            |



**Vehicle Platform:**  
**ECU Name: Seat Zone Module (SZM)**  
**DTC Troubleshooting Data**

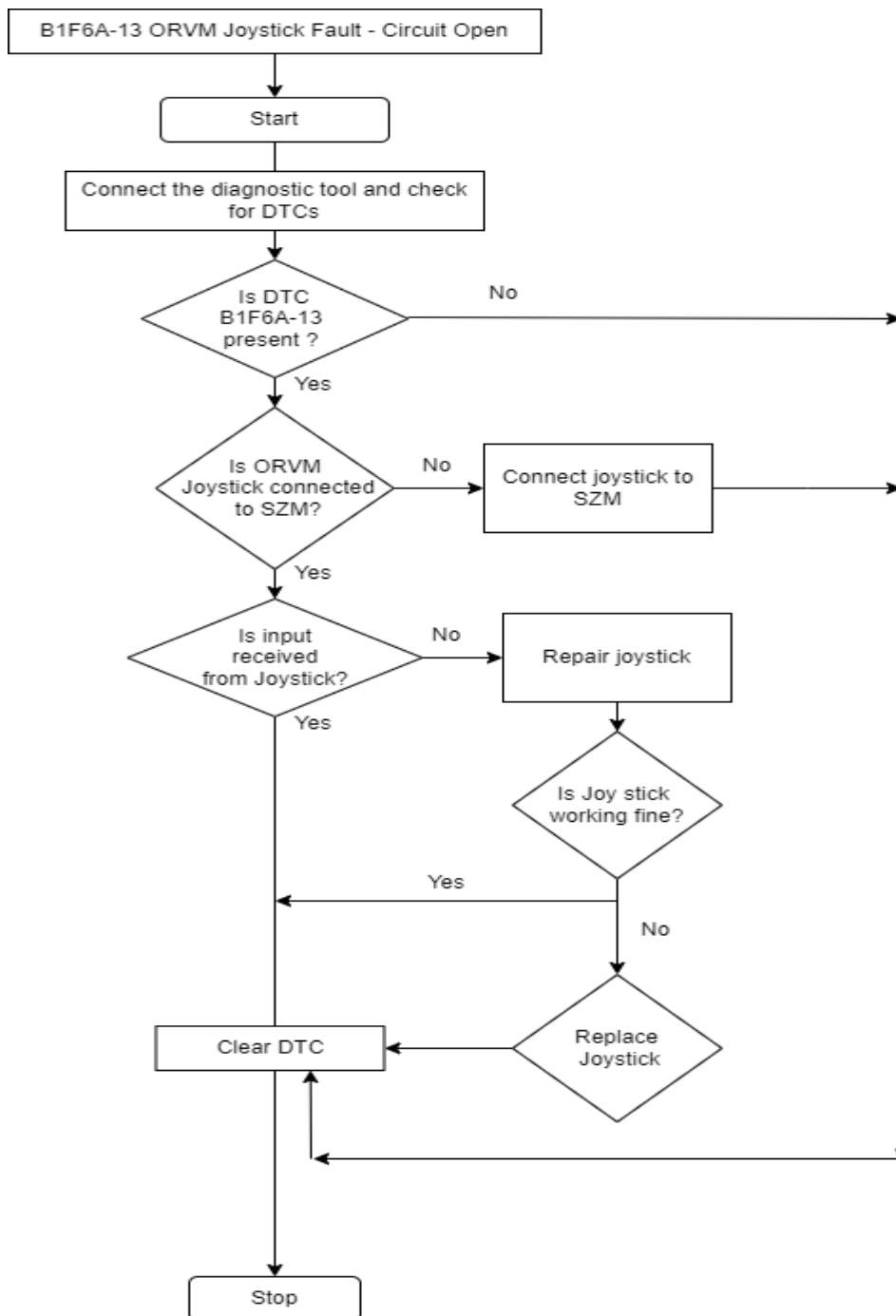
Version: 1.4

Date: 03-05-24

Text, Node 47

**Trouble Shooting: -**

Image, Node 47



|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

Text, Node 47

**DTC's confirmation: -**

After rectification, ensure the following points using TML diagnostic tool before handing over to customer.

- a. DTC should not be present in ECU memory.

**Circuit Schematic Diagram: -**

N.A

**Component Location & Image: -**

N.A

**Connector Location: -**

N.A

**Connector View & Information: -**

N.A

**Inspection Method: -**

N.A

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

**APPENDIX:**

| Sr no<br>(DTCId) | DisplayDTCCode | DTC Description                                                                                                                               | Page No | Mask<br>status |
|------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|
| 1                | U0028-13       | Vehicle Communication Bus A open circuit                                                                                                      | 8       | N              |
| 2                | B155A-01       | General Electric Failure General Electrical Failure of the motor or its Hall effect sensor failure or sensor's harness issues. (Slide Motor)  | 11      | N              |
| 3                | B155A-11       | Slide Motor Output Circuit Short to Ground                                                                                                    | 14      | N              |
| 4                | B155A-12       | Slide Motor Output Circuit Short to Battery                                                                                                   | 17      | N              |
| 5                | B155A-18       | Slide Motor Output Open Circuit                                                                                                               | 20      | N              |
| 6                | B155A-19       | Slide Motor Output Circuit Current Above Threshold                                                                                            | 23      | N              |
| 7                | B155B-01       | General Electrical Failure of the motor or its Hall effect sensor failure or sensor's harness issues. (Back rest Motor)                       | 26      | N              |
| 8                | B155B-11       | Back Rest Motor Output Circuit Short to Ground                                                                                                | 29      | N              |
| 9                | B155B-12       | Back Rest Motor Output Circuit Short to Battery                                                                                               | 32      | N              |
| 10               | B155B-18       | Back Rest Motor Output Open Circuit                                                                                                           | 35      | N              |
| 11               | B155B-19       | Backrest Motor Output Circuit Current Above Threshold                                                                                         | 38      | N              |
| 12               | B1561-01       | General Electric Failure General Electrical Failure of the motor or its Hall effect sensor failure or sensor's harness issues. (Height Motor) | 41      | N              |
| 13               | B1561-11       | Height Motor Output Circuit Short to Ground                                                                                                   | 44      | N              |
| 14               | B1561-12       | Height Motor Output Circuit Short to Battery                                                                                                  | 47      | N              |
| 15               | B1561-18       | Height Motor Output Open Circuit                                                                                                              | 50      | N              |
| 16               | B1561-19       | Height Motor Output Circuit Current Above Threshold                                                                                           | 53      | N              |
| 17               | B1D1E-24       | Switch Pack Switch stuck high                                                                                                                 | 56      | N              |
| 18               | B1D1E-87       | Seat Switch Pack Missing Message                                                                                                              | 59      | N              |
| 19               | U1000-00       | Solid State Driver Protection Active - Driver Disabled (Output driver is disabled temporarily)                                                | 63      | N              |
| 20               | U3000-93       | Control Module-SZM control module's Application software and Calibration are not matching.                                                    | 66      | N              |

|                 |                  |
|-----------------|------------------|
| Author: BCM COC | Page: 150 of 151 |
|-----------------|------------------|

|                                                                                  |                                                                                                        |                |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------|
|  | <b>Vehicle Platform:</b><br><b>ECU Name: Seat Zone Module (SZM)</b><br><b>DTC Troubleshooting Data</b> | Version: 1.4   |
|                                                                                  |                                                                                                        | Date: 03-05-24 |

|    |            |                                                                   |     |   |
|----|------------|-------------------------------------------------------------------|-----|---|
| 21 | U3000-49   | Control Module-Output driver is disabled permanently              | 68  | N |
| 22 | B160F-19   | LIN Power Feed-Circuit Current Above Threshold                    | 70  | N |
| 23 | B2233 - 13 | ORVM Co-Driver horizontal axis motor Open Circuit                 | 73  | N |
| 24 | B2233-11   | ORVM Co-Driver horizontal axis motor STG                          | 76  | N |
| 25 | B2233-12   | ORVM Co-Driver horizontal axis motor STB                          | 79  | N |
| 26 | B2233-19   | ORVM Co-Driver horizontal axis motor Over current (motor failure) | 82  | N |
| 27 | B2233 - 01 | ORVM co-driver horizontal axis motor- General feedback failure    | 85  | N |
| 28 | B2233 - 18 | ORVM Co-Driver horizontal axis motor under current                | 88  | N |
| 29 | B2231-13   | ORVM Co-Driver vertical axis motor Open Circuit                   | 91  | N |
| 30 | B2231-11   | ORVM Co-Driver vertical axis motor STG                            | 94  | N |
| 31 | B2231-12   | ORVM Co-Driver vertical axis motor STB                            | 97  | N |
| 32 | B2231-19   | ORVM Co-Driver vertical axis motor Over current (motor failure)   | 100 | N |
| 33 | B2231-01   | ORVM Co-Driver Vertical Axis Motor- general feedback failure      | 103 | N |
| 34 | B2231 - 18 | ORVM Co-Driver vertical axis motor under current                  | 106 | N |
| 35 | B2207 - 13 | ORVM Driver horizontal axis motor Open Circuit                    | 109 | N |
| 36 | B2207 - 11 | ORVM Driver horizontal axis motor STG                             | 112 | N |
| 37 | B2207 - 12 | ORVM Driver horizontal axis motor STB                             | 115 | N |
| 38 | B2207-19   | ORVM Driver horizontal axis motor Over current (motor failure)    | 118 | N |
| 39 | B2207 - 01 | driver horizontal axis motor- general feedback failure            | 121 | N |
| 40 | B2207 - 18 | ORVM Driver horizontal axis motor under current                   | 124 | N |
| 41 | B2203 - 13 | ORVM Driver vertical axis motor Open Circuit                      | 127 | N |
| 42 | B1F6B-11   | SZM ORVM Driver vertical axis motor STG                           | 130 | N |
| 43 | B2203 - 15 | ORVM Driver vertical axis motor STB                               | 133 | N |
| 44 | B2203 - 19 | ORVM Driver vertical axis motor Over current (motor failure)      | 136 | N |
| 45 | B2203 - 01 | driver vertical axis motor- general feedback failure              | 139 | N |
| 46 | B2203 - 18 | ORVM Driver vertical axis motor under current                     | 142 | N |
| 47 | B1F6A-71   | ORVM Joystick stuck - Actuator stuck                              | 145 | N |
| 48 | B1F6A-13   | ORVM Joystick Fault – Circuit Open                                | 147 | N |